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Master's thesis

NOVEL RATING SYSTEM FOR CHESS

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Chapter 1

Introduction

1.1 Background

Chess ratings are one of the most visible examples of a numerical skill model used in a real competitive environment. A single number is expected to summarize the historical strength of a player, predict future results, seed tournaments, support title norms, and provide an intuitive public ranking. The historical popularity of Elo-type ratings comes from their simplicity: after a game, the two players' ratings are updated according to the difference between the observed score and the expected score implied by the rating difference [15]. This idea is closely related to a much broader family of paired-comparison models, whose origins include the works of Zermelo, Thurstone, Mosteller, and Bradley–Terry [8, 9, 10, 11].

The practical success of Elo-style systems does not mean that the modelling assumptions are fully realistic. A classical rating system treats all players as if they belonged to one comparable global population. In reality, the set of games is sparse, time-dependent, and highly structured. Players usually do not face a uniformly random sample of the world. They mostly play in tournaments accessible to them, in their federation, in neighbouring federations, in age-category events, in school and junior circuits, or in elite closed tournaments. The resulting dataset is not only a sequence of game outcomes, but also a large interaction network. Community detection methods such as modularity maximisation and the Louvain algorithm were developed precisely for such networks, where dense local connectivity and sparse global bridges can reveal meaningful communities [37, 38, 39, 43, 41].

This thesis starts from the hypothesis that such structure matters for chess ratings. If two players have the same numerical rating, but one mostly plays in a local junior circuit and the other mostly plays in a strong open tournament circuit, the rating number alone may not represent the same predictive meaning. Similarly, a federation with many internally rated players and limited international cross-play can temporarily drift relative to another federation. A single global update rule can eventually correct such differences, but it may react slowly. The proposed approach therefore adds a dynamic pool-aware correction to an Elo-style rating system. The pools are not manually defined by federation. They are learned from the historical player interaction graph and are then used as latent context variables.

1.2 Problem statement

The problem investigated in the thesis is the following: can a rating system that uses latent player pools discovered from historical game interactions predict future chess results more accurately than a classical Elo-style baseline and than official FIDE ratings on the shared subset of games?

The work focuses on the standard chess rating context. The input is a chronological collection of games with player identifiers, event information, observed score, colours when available, rating-list information, and rating values. The output is not a public ranking proposal ready to replace FIDE. Instead, the goal is an experimental rating model that is evaluated by prediction quality. The central metric is mean squared error (MSE) of the predicted expected score. This is consistent with the interpretation of a game result as a score in $\{0, 0.5, 1\}$, rather than as a pure win/loss event.

Three difficulties make the problem interesting. First, the data are duplicated and incomplete in natural ways because the FIDE calculation pages describe games from each player’s perspective. The same physical game can appear twice and the displayed opponent rating can be affected by rating-difference rules. Second, the rating system must be evaluated chronologically. Future games must not influence past predictions, and the final test period must remain separate from parameter tuning. Third, the pool-aware correction must be useful out-of-sample. A model can easily fit historical communities but still fail to generalize if it overfits particular pools or pairwise interactions.

1.3 Aim and scope of the thesis

The main aim is to design, implement, and evaluate a dynamic pool-aware rating system for chess players. The work is empirical: it combines classical rating ideas, paired-comparison modelling, graph community detection, and chronological validation in one experimental system.

The scope of the work includes:

- collecting and transforming historical FIDE game-calculation data into a unique-game dataset;
- recovering rating information when displayed rating differences are affected by regulation-specific caps;
- constructing a player interaction graph and discovering latent pools before the main evaluation period;
- defining a rating model with base rating, uncertainty, provisional entry, dynamic pool exposure, and optional pool-pair interactions;
- selecting parameters only on the validation period;
- evaluating the selected model on a held-out final test period;
- performing ablation, sensitivity, calibration, stress-test, and explainability experiments.

A secondary aim is to inspect the learned pool structure. Since pools are latent variables inferred from the player interaction graph, their composition is analysed separately using federation, age, elite-player and cross-pool interaction data. This supplementary analysis checks whether the learned pools correspond to recognisable chess environments.

1.4 Structure of the thesis

Chapter 2 introduces the main concepts used in the analysis, including expected score, paired comparisons, uncertainty, interaction graphs, community detection and chronological validation. Chapter 3 reviews prior work on rating systems, rating-pool effects, dynamic ratings, graph community detection and probabilistic evaluation. Chapter 4 describes the data source, experimental split and evaluation metrics, while Chapter 5 explains how public FIDE calculation records are reconstructed into a chronological unique-game dataset.

Chapter 6 presents the pool-aware rating model, and Chapter 7 describes parameter search and model selection. The main predictive results are reported in Chapter 8. Chapter 9 contains sensitivity, calibration and robustness experiments. Chapter 10 discusses interpretation, deployment constraints and practical implications. Chapter 11 concludes the work. A supplementary analysis of latent player pools, including federation, age, elite-player and cross-pool interaction data, is provided in Appendix A.

Chapter 2

Basic Concepts

2.1 Expected score

In chess rating systems the predicted output of a game is usually an expected score rather than a direct win probability. A player receives score 1 for a win, 0.5 for a draw, and 0 for a loss. If S_A denotes the score of player A , then the rating model estimates

$$E_A = \mathbb{E}[S_A].$$

For a deterministic game outcome this is also the probability-weighted average

$$E_A = P(A \text{ wins}) + 0.5P(\text{draw}).$$

The same expected score can correspond to different win/draw/loss decompositions. For example, an expected score of 0.70 can come from

$$P(A \text{ wins}) = 0.65, \quad P(\text{draw}) = 0.10, \quad P(A \text{ loses}) = 0.25,$$

because

$$0.65 + 0.5 \cdot 0.10 = 0.70.$$

The same expected score can also come from a more draw-heavy forecast, for example

$$P(A \text{ wins}) = 0.45, \quad P(\text{draw}) = 0.50, \quad P(A \text{ loses}) = 0.05,$$

because

$$0.45 + 0.5 \cdot 0.50 = 0.70.$$

The experiments evaluate expected score because it is directly compatible with Elo-style updates and with the MSE objective.

2.2 Classical Elo expectation

The classical Elo system maps a rating difference into an expected score by a logistic curve [15]. If R_A and R_B are ratings and $D = R_A - R_B$, then the expected score of player A is commonly written as

$$E_A = \frac{1}{1 + 10^{-D/400}}.$$

The parameter 400 controls the steepness of the curve. A rating difference of 400 points corresponds to an expected score of approximately 0.91. In the proposed model the same general logistic idea is

kept, but the effective rating difference can include pool corrections, white advantage, uncertainty, and other dynamic components.

After a game, a simple Elo update is

$$R'_A = R_A + K(S_A - E_A),$$

where K controls the update speed. A large value of K allows rapid adaptation but increases volatility. A small value of K makes ratings stable but slow to correct. Official rating systems must balance both goals. The FIDE regulations define several practical coefficients and procedures around the Elo concept [32, 36].

2.3 Paired comparisons

Elo ratings are part of a larger class of paired-comparison models. In these models the observation is a comparison between two objects, players, teams, or alternatives. The unobserved strength of each object is represented by a latent parameter. Classical examples include the Thurstone–Mosteller model and the Bradley–Terry model [9, 10, 11]. Extensions add ties, order effects, and other corrections [12, 13, 14, 31].

A chess game is a natural paired comparison, but with two complications. First, draws are frequent and are not merely random noise. Second, the comparison graph is sparse and time-dependent. Many players never play each other, while local groups can be extremely dense. This means that a purely transitive one-dimensional rating scale may not fully capture all predictive structure.

2.4 Rating uncertainty

A rating number is not equally reliable for every player. A player with hundreds of recent games is estimated more precisely than a new player with five games. Dynamic rating systems such as Glicko and TrueSkill explicitly model uncertainty [17, 18, 19, 20]. In a practical rating system uncertainty can affect both updates and predictions. A player with high uncertainty may need a higher effective K so that the system can adapt quickly. The predictive curve may also become flatter when either player is uncertain.

The proposed system uses uncertainty in a lightweight way. It maintains a player uncertainty state, updates it after events, grows it during inactivity, and uses it to adjust both rating updates and expected-score scale. This is not a full Bayesian posterior like TrueSkill, but it follows the same intuition that player strength estimates should not all have equal precision.

2.5 Player interaction graph

The historical set of games can be viewed as a graph. Vertices are players. An edge between two vertices indicates that the players have faced each other, possibly with edge weight equal to the number of games. Such a graph captures information that is absent from a simple list of ratings. For example, two players may have similar ratings but belong to very different parts of the graph.

Network analysis provides tools for studying such structures [48, 40]. In this thesis the graph is used primarily for community detection. The aim is not to manually encode federations or countries. Instead, the system discovers groups of players who were historically connected by games before the main evaluation period.

2.6 Community detection

Community detection seeks groups of vertices that are more densely connected internally than externally. Modularity is a common quality function for this task [38, 39]. The Louvain method is a fast heuristic for modularity optimisation and is widely used for large graphs [43]. Later work showed that Louvain communities can be disconnected in some cases, and the Leiden algorithm was proposed to improve community quality [44].

This thesis uses Louvain-style community detection to build latent player pools. The choice is pragmatic: the graph is large, the pools need to be computed once before the evaluation period, and the method is fast enough for repeated experimentation. The discovered communities are later analysed to check whether they correspond to meaningful chess environments.

2.7 Chronological validation

In rating-system evaluation, time order is essential. A model that uses future games to tune past ratings gives an unrealistic estimate of prediction quality. Therefore the data are split chronologically. Historical games before the main evaluation period are used for pool discovery and replay. A training period and validation period are used to tune parameters. The final test period is held out and used only after parameter selection.

This setup is closer to forecasting evaluation than to random cross-validation. Standard random folds can leak future information into the training process, especially in time-dependent data. The methodological literature on prediction assessment and time-series validation emphasises this distinction [57, 58, 59, 60]. The experiments follow the chronological approach because rating systems are sequential by nature.

Chapter 3

Literature Review

3.1 Classical paired-comparison models

The mathematical problem behind chess ratings predates the Elo system. Zermelo proposed a probabilistic ranking model for tournament results in 1929 [8]. Thurstone introduced a law of comparative judgment, modelling choices as comparisons of latent psychological quantities [9]. Mosteller developed a related paired-comparison framework, and Bradley and Terry proposed the model that later became one of the standard references in the field [10, 11].

These models share a common assumption: observed comparisons are generated by latent strengths. In binary comparisons, the probability that item i beats item j depends on the difference between their latent parameters. This difference may be transformed by a logistic function, a normal cumulative distribution function, or another monotonic link. Chess ratings inherit this paired-comparison structure, even when practical rating systems avoid explicit maximum-likelihood estimation over the entire player set.

Draws are particularly relevant in chess. Davidson proposed an extension of the Bradley–Terry model that accommodates ties [12]. Henery and Joe considered paired-comparison models in chess contexts [25, 26]. These works indicate that chess results should not be reduced mechanically to binary outcomes. Draws, colour advantage, rating range and player strength all affect observed scores.

Modern reviews describe paired-comparison modelling as a broad statistical field with many variants [13, 31, 14]. The rating model developed later in the work keeps the online update structure of Elo rather than estimating a full Bradley–Terry model over all players. Concepts from paired-comparison theory remain central: latent strength, expected score, colour effects, uncertainty and calibration.

3.2 Elo and official chess ratings

Elo’s rating system transformed statistical paired-comparison ideas into a practical algorithm for chess federations [15]. Its main advantage is operational simplicity. A rating list can be updated after a tournament or rating period without solving a large global optimisation problem. FIDE ratings follow this tradition and are published through official rating lists and the ratings database [33, 34].

The official FIDE regulations specify practical details such as rating lists, entry rules, development coefficients, expected-score tables and special treatment of large rating differences [32]. The 2024 rating reform and later amendments to the 400-point rule show that public rating systems are periodically adjusted in response to fairness, inflation, deflation and edge-case concerns [36, 35].

Official ratings serve more than a predictive function. They are used for tournament eligibility, title norms, pairings, regulations and public comparison between players. They must therefore remain transparent, stable and accepted by the chess community. Predictive modelling can use more complex mechanisms, but a public rating list imposes additional requirements that are not captured by MSE alone.

3.3 Rating pools, scale drift and rating comparability

The term *rating pool* has prior use in chess-rating discussions. Elo-type ratings are relative to the population of players and games from which they are estimated. If two populations interact only weakly, their numerical ratings may become difficult to compare even when the same update formula is used. This issue appears in discussions of rating inflation, deflation, pool expansion, federation-specific ratings, online ratings, historical comparisons and the entry of new players into established lists.

Glickman and Jones analysed the behaviour of the United States Chess Federation rating system and explicitly discussed changes in the established rating pool [1]. Their analysis distinguishes between changes observed for continuing players and changes caused by the changing composition of the active population. This distinction is relevant in chess because players continuously enter, leave and re-enter the rating system.

Sonas discussed related issues in analyses of rating inflation and deflation. His ChessBase article on rating inflation considered possible causes such as the number of players in the rating pool and the procedure used for initial ratings [2]. His later proposal for repairing the FIDE Standard Elo rating system argued that the standard rating distribution had become deflated and excessively spread out, and recommended compression and calculation changes [3]. The subsequent FIDE reform illustrates that public rating systems are adjusted when the behaviour of the rating pool is considered problematic [36, 32].

Historical rating systems address a related but distinct problem. Chessmetrics, developed by Sonas, provides retrospective ratings intended to compare players across historical periods and to reduce distortions caused by differences between rating eras [4, 5]. Chessmetrics is primarily a historical reconstruction system, whereas the experiments in later chapters use chronological prediction on modern FIDE game data.

Population-level models have also studied the evolution of chess rating distributions. Fenner, Levene and Loizou proposed a stochastic evolutionary model for chess ratings, explicitly modelling the arrival of players into the rating pool, the games they play and the effect of those games on the distribution of ratings [6]. Their work treats the rating pool as an evolving population and studies distributional consequences.

Other rating-system research improves Elo without explicitly modelling rating pools. Elo++ was developed for the Kaggle competition *Chess Ratings – Elo versus the Rest of the World* and used regularisation and cross-validation to improve predictive accuracy over an Elo baseline [28, 27]. Whole-History Rating estimates time-varying player strengths using a dynamic Bradley–Terry model and maximum a posteriori estimation over the whole rating history [7]. These systems provide relevant background for rating prediction, temporal strength estimation and regularisation.

The approach developed in later chapters uses a graph-derived version of the rating-pool idea. Communities are inferred from historical player interactions rather than assigned from federations or geography. These communities are then represented in an Elo-like chronological replay through pool exposure, pool offsets and pool-pair interactions. The term *latent pool* is used for these

graph-derived communities to distinguish them from official federations, rating lists or manually defined regions.

3.4 Dynamic rating systems

Several rating systems extend Elo by modelling uncertainty or skill drift. Glickman’s work is central in this area. The Glicko system and related dynamic paired-comparison models maintain both a rating and a rating deviation, allowing the system to represent uncertainty about a player [16, 17, 18]. Glicko-2 further adds volatility [19]. These systems are especially relevant when players have irregular activity patterns.

TrueSkill is another influential system. It was designed for multiplayer and team-based online games, but its Bayesian formulation is relevant to rating systems that maintain uncertainty over latent skill [20]. TrueSkill through time applied related ideas to the history of chess, emphasising the dynamic nature of player strength [21]. Approximate inference methods such as expectation propagation provide the computational background for such models [22]. Broader probabilistic machine learning texts place these ideas within latent-variable modelling, uncertainty and approximate inference [23, 24].

The model used in the experiments is not a full Bayesian rating model and does not maintain a complete posterior distribution over all players. It uses a deterministic online replay with additional state variables. The practical motivation is similar to dynamic rating systems: predictions should account for uncertainty, activity patterns and temporal change. The additional contextual dimension considered here is exposure to latent interaction pools.

3.5 Chess-rating competitions and prediction studies

The Kaggle competition *Chess Ratings – Elo versus the Rest of the World* asked participants to predict chess game outcomes more accurately than an Elo baseline [28]. The winning approach, Elo++, used regularisation and careful cross-validation to avoid overfitting [27]. The competition demonstrated that predictive improvements over Elo are possible and that evaluation protocols can strongly affect conclusions.

Sports forecasting studies have used Elo-style ratings outside chess. Football prediction work shows that rating systems can be competitive components of match-result forecasts [29, 30]. These applications support the interpretation of rating systems as sequential prediction mechanisms rather than only ranking tables.

A recurring issue in such work is temporal evaluation. A model that performs well on randomly mixed games may fail when forecasting future games. For this reason, the experiments use chronological train, validation and test periods. Aggregate performance is supplemented by monthly stability, calibration and stress-test analyses.

3.6 Graph community detection and latent pools

The second methodological foundation is graph community detection. In social and biological networks, communities are groups of vertices with dense internal connections and sparser external connections [37, 38]. Modularity is a widely used objective for identifying such communities [39]. The Louvain method made modularity optimisation feasible at large scale [43]. Later surveys describe families of methods and known limitations, including resolution effects and ambiguity in community definitions [41, 42].

The Leiden method addresses some weaknesses of Louvain by improving the connectedness of discovered communities [44]. Other approaches use random walks or information-theoretic criteria to find communities at different scales [45, 46, 47]. Social network analysis provides the conceptual language for interpreting such structures [48, 40]. Heterogeneous and scale-free network phenomena further indicate that large interaction graphs rarely behave like uniform random graphs [49].

In the chess setting, a player interaction graph is formed by games between players. Communities in such a graph may reflect federations, regions, tournament circuits, junior ecosystems or elite events. The model does not assign these meanings in advance. Latent pools are discovered from historical interactions and later examined through predictive tests and descriptive analyses.

3.7 Scoring rules and calibration

The experiments evaluate predicted expected scores using mean squared error. This is closely related to the Brier score, introduced for probabilistic forecasts [50]. Proper scoring rules provide a principled framework for evaluating probabilistic predictions: a forecaster is rewarded for reporting probabilities that correspond to empirical frequencies [53]. The prequential approach emphasises sequential prediction and evaluation, which matches the rating-system setting [52].

Calibration is distinct from discrimination. A model may correctly rank favourites and underdogs while still producing probabilities that are too extreme. Forecast evaluation literature discusses calibration, sharpness and decompositions of probability scores [51, 54]. Machine-learning work has also studied probability calibration for supervised classifiers and neural networks [55, 56]. The calibration-bin experiment checks whether predicted expected scores align with empirical outcomes in the final test period.

3.8 Model selection and evaluation protocol

Rating models with many interacting components cannot be assessed only by in-sample fit. Repeated tuning on the same period can select parameters that exploit accidental properties of that period. This problem is well known in model selection and hyperparameter optimisation [57, 58, 63, 64, 65].

The experimental protocol separates three temporal roles. Older data initialise the system and define latent pools. A validation period is used for parameter selection. A later test period is reserved for the final comparison. This chronological design follows the logic of sequential forecasting and avoids random folds that would mix future and past games [52, 59, 60]. Implementation details are discussed only where they affect this protocol: reciprocal rating recovery, duplicated game reconstruction, pool discovery before evaluation and chronological replay.

Chapter 4

Materials and Methods

4.1 Data source

The empirical work is based on FIDE standard-rating game-calculation data. FIDE publishes official rating information and top lists through its ratings service [33, 34]. The underlying rating regulations describe how scores, rating differences, rating periods and coefficients are handled [32]. The thesis uses the data as a historical source of rated games, player identifiers, rating-list values, event information, colours and results.

A practical challenge is that the public calculation pages are player-centric. A physical game can be visible from the perspective of both players. Therefore the raw data are not yet a clean game table. They are directed records: player A 's page contains a row against player B , and player B 's page can contain the reciprocal row. The data preparation pipeline must reconcile these records and build a unique-game representation.

The dataset used for the final experiments contains 24,771,594 replayed game records in the cache, 597,530 players and 57 latent pools. The final test period contains 7,393,544 unique games. These quantities make the problem large enough that evaluation speed matters, but still small enough for carefully engineered local computation.

4.2 Temporal split

The experiments use chronological splits. Pool discovery is performed using historical interactions before the main evaluation period. Parameter search uses training and validation periods. The final test period begins in January 2022 and is not used for parameter tuning.

The key split points are:

- pool-discovery period: games up to December 2014;
- training period for parameter search: January 2015 to December 2018;
- validation period: January 2019 to December 2021;
- final test period: January 2022 to the end of the dataset.

This setup is deliberately conservative. The learned pools are based only on the older interaction graph. The model parameters are selected only by validation quality. The final test period is used once to report performance. The same idea is standard in forecasting evaluation, where future observations must not influence the model-selection process [52, 59, 60].

4.3 Evaluation metrics

The primary metric is mean squared error:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (S_i - \hat{E}_i)^2,$$

where S_i is the observed score of the first player in game i and $\hat{E}_i$ is the predicted expected score before observing the game. The secondary metric is mean absolute error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |S_i - \hat{E}_i|.$$

MSE is emphasised because it is a strictly proper scoring-rule analogue for expected-score forecasts and penalises overconfident large mistakes more strongly [50, 53]. The official FIDE rating can also be converted into an expected score on games where both ratings are available. This allows a shared-subset comparison.

The experiments report:

- all-game MSE and MAE;
- MSE on the subset shared with official FIDE ratings;
- official FIDE MSE on the same shared subset;
- improvement relative to classical Elo;
- improvement relative to official FIDE on the shared subset;
- monthly stability metrics;
- calibration error.

4.4 Model comparison philosophy

The purpose of the comparison is not to claim that the proposed system is simpler or more transparent than official FIDE ratings. Official ratings are constrained by governance, stability and public interpretability. The proposed model is an experimental predictive system. Therefore its success criterion is predictive accuracy, not immediate public deployability.

The baseline is a classical Elo-style system replayed on the same unique-game sequence. Additional variants remove components from the proposed model. This ablation strategy is important because the model contains several mechanisms. A single final number would not reveal which mechanism matters. The final chapters therefore report both the aggregate test comparison and component-level analyses.

4.5 Threats to validity

The main threats to validity are the following. First, the dataset is reconstructed from public FIDE calculation information and can inherit missingness or formatting limitations. Second, the exact meaning of some metadata fields can vary across years. Third, latent pools are discovered from a historical graph and may change over time.

These limitations do not invalidate the predictive experiment, but they affect interpretation. For example, a wider effective rating range can improve prediction while still the public usage

might require a later normalisation step to avoid confusion. Similarly, a pool that aligns with a federation should not automatically be interpreted as a national-strength statement; it is first an internal predictive correction discovered from a graph.

Chapter 5

Data Preparation

5.1 Overview

The experiments require a chronological table of unique rated games. Each record should represent one physical game and contain two player identifiers, the result, rating-period information, event metadata, colour information when available, and official ratings needed for comparison. The public FIDE calculation data are not organised directly in this form. They are player-centric rating-calculation records: a game may appear once from the perspective of each player, and some values reflect the way a rating change was calculated rather than a neutral game archive.

The preparation pipeline converts these directed calculation rows into a unique-game dataset. It consists of four main stages: collection of directed rows, conversion to a columnar intermediate representation, recovery of opponent-rating information, and construction of unique game records. The purpose is not to reproduce the internal FIDE database. The purpose is to obtain a consistent chronological sequence suitable for replaying rating systems under the same experimental protocol.

If an official game-level export with stable identifiers were available, the reconstruction stage would be simpler. The experimental requirements, however, would remain the same: each replayed game must have a chronological position, two player identifiers, a result, colour information when available, event grouping and rating information separated from any values modified for rating-calculation purposes.

5.2 Directed calculation rows

The first stage collects calculation rows from publicly available FIDE rating information [33]. Each row describes a game or rating-calculation entry from the perspective of one player. Such rows include the player’s identifier, opponent information, event metadata, score, colour if available, rating period and displayed ratings.

The collection stage is conservative. Successful rows are stored, while problematic cases are left for later inspection instead of being silently guessed. This is important in a large historical dataset, where rare inconsistencies are expected. Later stages therefore prefer strict matching and avoid forcing ambiguous records into reciprocal pairs.

5.3 Columnar conversion

The collected rows are converted into a columnar intermediate representation. Rating experiments repeatedly scan a limited set of fields over millions of records, so a column-oriented layout is appropriate for chronological filtering, joining and replay preparation. The conversion stage

standardises data types, removes temporary or incomplete files from the processing stream, and records parsing errors separately.

This stage does not change the meaning of the data. It only turns heterogeneous collected rows into a stable representation that can be checked and processed consistently. Separating collection from conversion also makes later stages easier to inspect: errors caused by missing source rows, malformed records and reciprocal matching are not mixed into a single step.

5.4 Recovering opponent ratings

A specific difficulty concerns the opponent rating displayed in calculation records. Rating regulations may treat very large rating differences specially when calculating rating changes [32]. In such cases, the displayed or calculation-effective opponent rating may differ from the opponent’s actual rating at the time. Treating every displayed value as an actual rating would distort comparisons with official expected-score calculations.

The pipeline therefore attempts to recover the actual opponent rating from the reciprocal row. The matching uses player identifiers, opponent identifiers, event metadata, rating period, date information and score consistency. Strict reciprocal matches are preferred. Relaxed matching is used only when ambiguity is low. When a reliable match cannot be established, the record is not forced into a recovered pair.

This distinction matters for evaluation. The experimental model predicts game outcomes from its own internal state, while the official-rating comparison uses the subset of games for which the relevant FIDE rating information can be reconstructed consistently. Separating displayed calculation values from recovered actual opponent ratings prevents rating-cap effects from being treated as ordinary opponent ratings.

5.5 Unique-game construction

The next stage builds unique physical games. A reciprocal pair of directed rows should become one game. However, repeated games between the same players in the same event are legitimate, especially in match or round-robin contexts. Deduplication therefore cannot rely only on player identifiers and event names. It must preserve distinct games while collapsing only true duplicates.

The resulting table contains one row per reconstructed game. Each row includes the two players, the score from one fixed perspective, colour sign, rating month, event grouping and available official ratings. This table is the basis for all rating replays. Predictions are made before state updates, and updates are applied chronologically with event-aware logic.

The unique-game construction is intentionally independent of the proposed rating model. It is not tuned to improve the model’s result. Its role is to define a stable evaluation sequence that can be used by the baseline Elo system, ablation variants and the full model under the same conditions.

5.6 Latent pool discovery dataset

Latent pools are derived from a historical interaction graph. The graph is built only from games before the main evaluation period, so pool discovery does not use future test information. Vertices represent players and edges represent repeated interactions between them. Very weak edges can be filtered to reduce noise, and graph size is bounded for computational feasibility.

Community detection produces pool identifiers used later in the rating replay. Small or weakly supported communities are merged into the default pool. Players appearing after the pool-discovery

period can be assigned using observed neighbourhood votes when sufficient information is available. Otherwise they remain in the default pool until their interaction history becomes informative.

The final cache used in the experiments contains 57 pool identifiers. Pool 0 is the default or unassigned pool and contains a large fraction of all players. Non-zero pools represent the main latent interaction communities. Their interpretation is not assumed during model fitting; it is examined later through federation, age, junior, elite-player and cross-pool interaction analyses.

5.7 Data quality considerations

Large historical sports datasets are rarely perfectly clean, especially when reconstructed from public calculation views rather than received as a curated game-level export. The preparation pipeline therefore follows conservative rules. Ambiguous reciprocal matches are not forced. Rating values affected by calculation rules are separated from recovered actual ratings. Duplicated directed rows are collapsed only when they can be treated as the same physical game.

The central requirement is chronological consistency. A rating replay must not use future information when predicting a game. For that reason, pool discovery is restricted to historical data, parameter selection is separated from the final test period, and game reconstruction is completed before model evaluation. The preprocessing stages approximate the set of rated physical games as faithfully as possible under these constraints.

Chapter 6

Proposed Rating System

6.1 Classical baseline

The baseline is an Elo-style rating system replayed on the same unique-game sequence. It maintains one rating per player and predicts expected score from the rating difference. Its role is methodological: it provides a simple reference controlled by the same data split and the same game order. The baseline should not be confused with official FIDE ratings, which include additional historical procedures, rating periods and regulations [32].

6.2 Effective rating

The proposed system maintains several state variables for each player. The central predictive quantity is an effective rating:

$$R_i^{\text{eff}}(t) = R_i^{\text{slow}}(t) + F_i(t) + w_{\text{pool}}P_i(t),$$

where $R_i^{\text{slow}}(t)$ is the long-term base rating, $F_i(t)$ is a short-term form component, $P_i(t)$ is the player's dynamic pool value, and w_{pool} controls the strength of the pool contribution.

The dynamic pool value is computed from the player's current exposure to latent pools:

$$P_i(t) = \sum_{p \in \mathcal{A}_i(t)} e_{i,p}(t)b_p(t),$$

where $\mathcal{A}_i(t)$ is the set of active pools stored for player i , $e_{i,p}(t)$ is the exposure of player i to pool p , and $b_p(t)$ is the learned offset of pool p at time t .

For a game between players A and B , the model forms an effective difference from the perspective of player A :

$$D_{AB}(t) = R_A^{\text{eff}}(t) - R_B^{\text{eff}}(t) + \Delta_{\text{pair}}(p_A, p_B, t) + \Delta_{\text{white}}.$$

The pairwise pool term is written as

$$\Delta_{\text{pair}}(p_A, p_B, t) = w_{\text{pair}}B_{p_A, p_B}(t),$$

where $B_{p_A, p_B}(t)$ is the learned interaction offset between the dominant pools of the two players and w_{pair} is its global weight. The colour term is

$$\Delta_{\text{white}} = c_A w_{\text{white}},$$

where $c_A = 1$ if player A has White, $c_A = -1$ if player A has Black, and $c_A = 0$ if colour is unavailable. The parameter w_{white} is the learned white-advantage correction.

The expected score is then computed by a logistic curve:

$$E_A(t) = \frac{1}{1 + 10^{-D_{AB}(t)/s_{AB}(t)}}.$$

In classical Elo, the scale is fixed at 400. In the proposed system, $s_{AB}(t)$ has a learned base value and can be adjusted by rating level, rating difference and player uncertainty.

The model remains Elo-like in its main structure: a rating difference is transformed into an expected score, and internal states are updated from residuals between observed and expected scores. The difference is that the rating difference is augmented with pool context, colour, uncertainty and event-level information.

6.3 Uncertainty

Each player has an uncertainty state $u_i(t)$. New or inactive players are more uncertain; active players with many recent games become more certain. Uncertainty affects both updates and predictions. It can increase the effective update size for less certain players and flatten the expected-score curve when uncertainty is high.

This mechanism is inspired by dynamic rating systems such as Glicko and TrueSkill [17, 18, 20], but the implementation is lighter and designed for fast chronological replay. The model uses an initial uncertainty, a lower bound, an upper bound, game-based decay and inactivity growth. The selected numerical values of these parameters are reported after the model-selection procedure in Section 7.4.

6.4 Provisional player entry

New players are difficult for rating systems. A fixed initial rating can be inaccurate for strong juniors or for experienced players entering the FIDE list later. The proposed system therefore estimates an early performance-based entry rating:

$$R_i^{\text{entry}} = (1 - \lambda_{\text{prov}})R_0 + \lambda_{\text{prov}}\hat{R}_i^{\text{perf}},$$

where R_0 is the prior entry rating, $\hat{R}_i^{\text{perf}}$ is an early performance estimate against rated opponents, and λ_{prov} is the provisional blend parameter. The performance estimate is clipped to avoid extreme initial ratings.

During the provisional period, rating updates can also be accelerated:

$$K_i^{\text{prov}} = m_{\text{prov}}K_{\text{base}},$$

where K_{base} is the base update coefficient and m_{prov} is the provisional update multiplier. The mechanism is sequential: only games already available at a given point in the replay can influence the player's state.

6.5 Latent pools

Latent pools are discovered from the historical player interaction graph using community detection. They are not assigned manually from federation, geography or title level. A pool is therefore a graph-derived competitive environment rather than an administrative label.

Each pool has a learned offset $b_p(t)$ used in the dynamic pool value $P_i(t)$. A player can have exposure to more than one pool. This allows the model to represent players whose competitive

history is mostly local, as well as players who gradually move into broader international or elite environments.

The interpretation of the learned pools is separated from their use in prediction. During model fitting, pools are numerical state components inferred from the historical graph. Their federation, regional, junior and elite-player structure is analysed separately in Appendix A.

6.6 Dynamic pool exposure

A static pool assignment would be too rigid. Players can move between tournament circuits, improve rapidly, change federation, or become internationally active. The system therefore maintains exposure weights $e_{i,p}(t)$ rather than assigning each player permanently to one community.

A player’s exposure vector is updated after events using residuals and opponent context. The exposure mechanism allows a player to be influenced by several environments while keeping the model compact. The maximum number of active exposures is controlled by a global parameter selected during validation.

This representation avoids treating a pool as a permanent identity. It is instead a component of the player’s recent and historical competitive context.

6.7 Pool-pair interactions

The model includes optional pairwise pool interactions:

$$\Delta_{\text{pair}}(p_A, p_B, t) = w_{\text{pair}} B_{p_A, p_B}(t).$$

These interactions account for systematic differences when players from two dominant pools meet. They are separate from the individual pool offsets $b_p(t)$ used in $P_i(t)$.

The pairwise term increases model complexity, so its empirical value is evaluated through ablation rather than assumed in the model definition.

6.8 Event-level update

Games are replayed chronologically. Within an event, predictions are made before applying the event’s residuals. This avoids using later games from the same event to predict earlier games in that event.

For a game g , the residual is

$$r_g = S_g - E_g,$$

where S_g is the observed score and E_g is the predicted expected score. Event residuals are aggregated before updating player and pool states. The slow rating update is controlled by the base coefficient K_{base} , while additional factors can depend on uncertainty and provisional status. Pool offsets, form terms and exposure values are updated through their own coefficients.

The update logic is therefore sequential but not purely row-by-row. This matters in chess because rating periods and tournaments consist of multiple games. Event-aware replay is a compromise between game-level online prediction and period-level official rating calculation.

Chapter 7

Parameter Search and Model Selection

7.1 Search space

The proposed model has parameters controlling rating update speed, expected-score scale, uncertainty, form, pool dynamics, pair interactions, provisional entry and event normalisation. The search space is constrained to plausible ranges. Some parameters are clipped to remain inside those ranges. This is why several sensitivity experiments show identical values for very large multipliers: the candidate reaches the allowed maximum.

The search does not estimate every player rating as a free parameter. Player ratings are generated by replaying the chronological game sequence. The optimised quantities are global behavioural parameters. This makes the model usable as a rating system, not only as an offline fitted matrix.

7.2 Validation-only selection

The selected parameter vector is chosen using validation performance only. The saved best-parameter file explicitly records that the test period was not used for tuning. The validation objective combines validation MSE, shared-subset MSE, year-balanced validation MSE, worst-year validation MSE and an overfitting penalty. This multi-part objective discourages parameters that perform well in one narrow period but generalize poorly.

Hyperparameter optimisation can overfit validation data, especially when many candidates are tried [63, 64, 65]. The final test period is therefore kept separate and is used only after the model has been selected.

7.3 Search procedure

The parameter search is evolutionary in structure. It starts from a set of seed candidates, extends the initial population with random candidates and mutated variants, evaluates candidates by chronological replay, and then repeatedly refines the best candidates. Evolutionary computing is suitable in this setting because the objective is non-convex, expensive to evaluate and affected by interactions between parameters [61, 62].

In the run used for the reported results, the fast search stage used a population of about 124 candidates and 100 generations. In each generation, duplicate candidates were removed before evaluation. The best candidates were sorted by validation fitness, and the top 28 candidates were retained as elites. The next generation was produced from these elites using four operators:

- differential mutation using three elite parents;
- crossover between two elite parents;
- Gaussian-style mutation of one elite parent;
- occasional random candidate generation.

The operators were sampled with fixed probabilities. Differential candidates were used for approximately one quarter of the new population when enough elites were available. Crossover and mutation produced most of the remaining candidates, while random candidates preserved exploration. All candidates were clipped to the allowed parameter ranges before evaluation.

The fast stage was followed by a full-cache refinement stage. The best candidates from the fast search were re-ranked on the complete training and validation replay. The refinement then used the best few candidates as parents and generated additional local mutations. In the reported run, 48 fast-stage finalists were carried forward, followed by 13 refinement rounds with approximately 78 candidates per round. A final coordinate-refinement stage tested positive and negative changes of individual parameters around the current best candidate. This stage used 3 rounds and about 83 coordinate candidates per round.

The search history contains approximately 9.5 thousand distinct candidates from the fast evolutionary stage and about 1.3 thousand candidates from full re-ranking, local refinement and coordinate refinement. A single candidate evaluation consists of a chronological replay of the rating system and calculation of validation metrics. The wall-clock time of one evaluation depends on the replay cache, CPU, parallel workers and whether the fast sample or full cache is used. In the full-cache diagnostic sweeps, one replay was typically on the order of tens of seconds on the machine used for the experiments, while fast-stage candidates were evaluated in parallel batches.

Figure 7.1 shows the improvement of validation fitness during the search. The progress is not perfectly smooth because candidate generation and evaluation include multiple objectives. The search is not claimed to find a global optimum. Its role is to find a strong validation-selected parameter vector under practical time constraints.

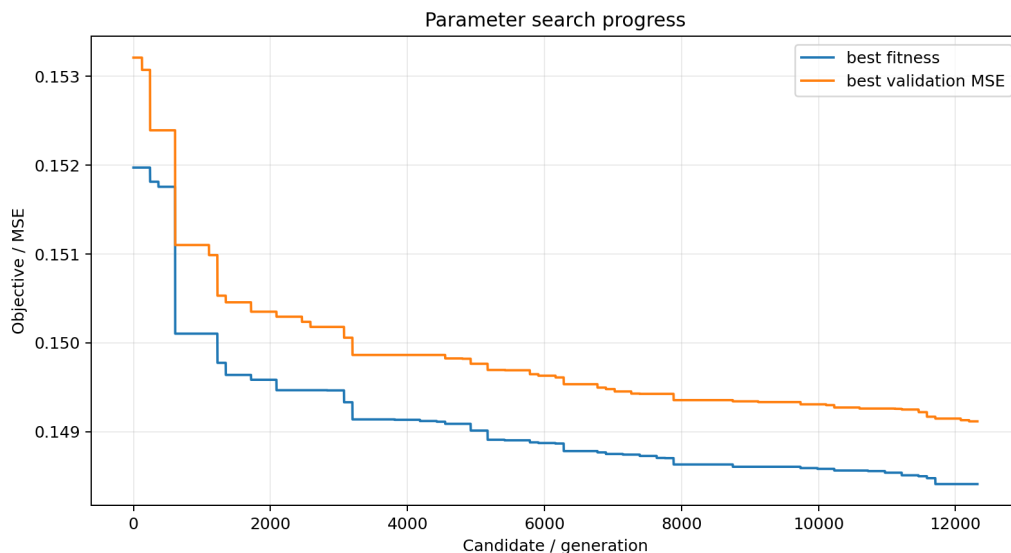


FIGURE 7.1: Progress of the parameter search. The plot shows the reduction of the validation objective during evolutionary refinement.

7.4 Final selected parameters

Table 7.1 reports the main parameters of the validation-selected model. The table is organised by model component rather than as a flat list of constants. This is necessary because only some parameters appear directly in the expected-score formula. Other parameters control the construction and update of the internal states used in prediction.

The main predictive equations from Chapter 6 are

$$R_i^{\text{eff}}(t) = R_i^{\text{slow}}(t) + F_i(t) + w_{\text{pool}}P_i(t),$$

$$P_i(t) = \sum_{p \in \mathcal{A}_i(t)} e_{i,p}(t)b_p(t),$$

$$D_{AB}(t) = R_A^{\text{eff}}(t) - R_B^{\text{eff}}(t) + w_{\text{pair}}B_{p_A,p_B}(t) + c_A w_{\text{white}},$$

and

$$E_A(t) = \frac{1}{1 + 10^{-D_{AB}(t)/s_{AB}(t)}}.$$

With this notation, w_{pool} is the pool-weight parameter, w_{pair} is the pool-pair weight, w_{white} is the white-advantage parameter, and the scale-base parameter contributes to $s_{AB}(t)$. The remaining parameters control how $R_i^{\text{slow}}(t)$, $F_i(t)$, $u_i(t)$, $b_p(t)$ and $e_{i,p}(t)$ are initialised or updated during chronological replay.

Model component	Parameter	Value	Mathematical role
Rating update	base K	37.540	Main coefficient in the update of $R_i^{\text{slow}}(t)$ from event residuals.
Expected-score scale	scale base	388.576	Base contribution to $s_{AB}(t)$ in $E_A(t) = (1 + 10^{-D_{AB}(t)/s_{AB}(t)})^{-1}$.
Colour correction	white advantage	36.106	This is w_{white} in the term $c_A w_{\text{white}}$ added to $D_{AB}(t)$.
Uncertainty	initial uncertainty	1.338	Initial value of the player uncertainty state $u_i(t)$.
Uncertainty	uncertainty floor	0.102	Lower bound for $u_i(t)$ after repeated activity.
Short-term form	form gain	0.396	Controls the update of the form term $F_i(t)$ from event residuals.
Pool update	pool K	4.337	Main coefficient in the update of pool offsets $b_p(t)$.
Pool contribution	pool weight	2.415	This is w_{pool} in $R_i^{\text{eff}}(t) = R_i^{\text{slow}}(t) + F_i(t) + w_{\text{pool}}P_i(t)$.
Pool exposure	pool exposure gain	0.009	Controls the update speed of exposure weights $e_{i,p}(t)$.
Pool exposure	initial home pool exposure	0.923	Initial value of $e_{i,p}(t)$ for the player's discovered or home pool.
Pool-pair interaction	pool pair weight	0.501	This is w_{pair} in $w_{\text{pair}}B_{p_A,p_B}(t)$ added to $D_{AB}(t)$.
Provisional entry	provisional games target	12.000	Number of early games used by the provisional-entry mechanism.
Provisional entry	provisional blend	0.724	This is λ_{prov} in $R_i^{\text{entry}} = (1 - \lambda_{\text{prov}})R_0 + \lambda_{\text{prov}}\widehat{R}_i^{\text{perf}}$.
Provisional entry	provisional K multiplier	1.804	This is m_{prov} in $K_i^{\text{prov}} = m_{\text{prov}}K_{\text{base}}$.
Pool exposure	max pool exposures	3.000	Maximum size of the active exposure set $\mathcal{A}_i(t)$.

TABLE 7.1: Selected parameters of the validation-chosen model and their mathematical roles in the rating system.

The selected values should be interpreted jointly. For example, the pool contribution depends on the product of the pool weight, learned pool offsets and player exposure values. Similarly, the expected score depends on both the effective difference $D_{AB}(t)$ and the scale $s_{AB}(t)$. The table gives the selected numerical configuration, while the importance of each component is evaluated later through ablation and sensitivity experiments.

7.5 Computational considerations

A full replay over millions of games is computationally expensive. The implementation uses a replay procedure capable of evaluating many candidate parameter vectors under the same chronological

protocol. The methodological requirement is consistency: every candidate must see the same past information at the same point in time. Implementation details are therefore reported only when they affect data reconstruction, pool discovery, parameter selection, or chronological evaluation.

Chapter 8

Main Experimental Results

8.1 Final test comparison

Table 8.1 reports the final comparison on the held-out test period. The proposed full model obtains test MSE 0.143993 and MAE 0.322149 over 7,393,544 games. The classical Elo baseline obtains MSE 0.156026 and MAE 0.347232. The relative MSE reduction is 7.71%.

The table also reports the shared-subset comparison with official FIDE ratings. This subset contains only games for which the official FIDE expected score can be reconstructed consistently. Official FIDE is therefore shown as a separate shared-subset baseline rather than as a repeated column for every model. On this subset, the proposed full model has MSE 0.139329, while official FIDE has MSE 0.149907.

Model	MSE	MAE	Shared MSE	vs. Elo [%]	vs. FIDE [%]
No form	0.143988	0.322156	0.139323	7.72	7.06
Proposed full	0.143993	0.322149	0.139329	7.71	7.06
No pair interaction	0.144050	0.322290	0.139339	7.68	7.05
No white advantage	0.144538	0.323217	0.139944	7.36	6.65
No uncertainty pred.	0.145920	0.317253	0.140316	6.48	6.40
Fixed 1500 entry	0.146356	0.324138	0.139669	6.20	6.83
Advanced seed	0.146868	0.325186	0.141038	5.87	5.92
Simplified	0.148872	0.318575	0.142179	4.59	5.16
Previous dynamic	0.150849	0.334480	0.144140	3.32	3.85
No dynamic exposure	0.151208	0.327758	0.141120	3.09	5.86
No performance entry	0.151462	0.322546	0.141042	2.93	5.91
9-game seed	0.154972	0.330937	0.141057	0.68	5.90
Classic Elo	0.156026	0.347232	0.146114	0.00	2.53
No pool	0.156241	0.331646	0.141211	-0.14	5.80
Official FIDE	—	—	0.149907	—	0.00

TABLE 8.1: Final test-period comparison of selected model variants. MSE and MAE are computed on all test games. Shared MSE is computed on the subset where the official FIDE expected score can also be reconstructed. Official FIDE is therefore reported as a separate shared-subset baseline. Improvements are expressed as MSE reductions.

The ranking of variants in Figure 8.1 gives the same picture. The full model and the variant without short-term form are almost indistinguishable. In contrast, the model without pools loses almost the entire advantage and becomes slightly worse than the classical baseline. Removing dynamic exposure or performance-based entry also causes large losses. These differences identify

the structural source of the improvement.

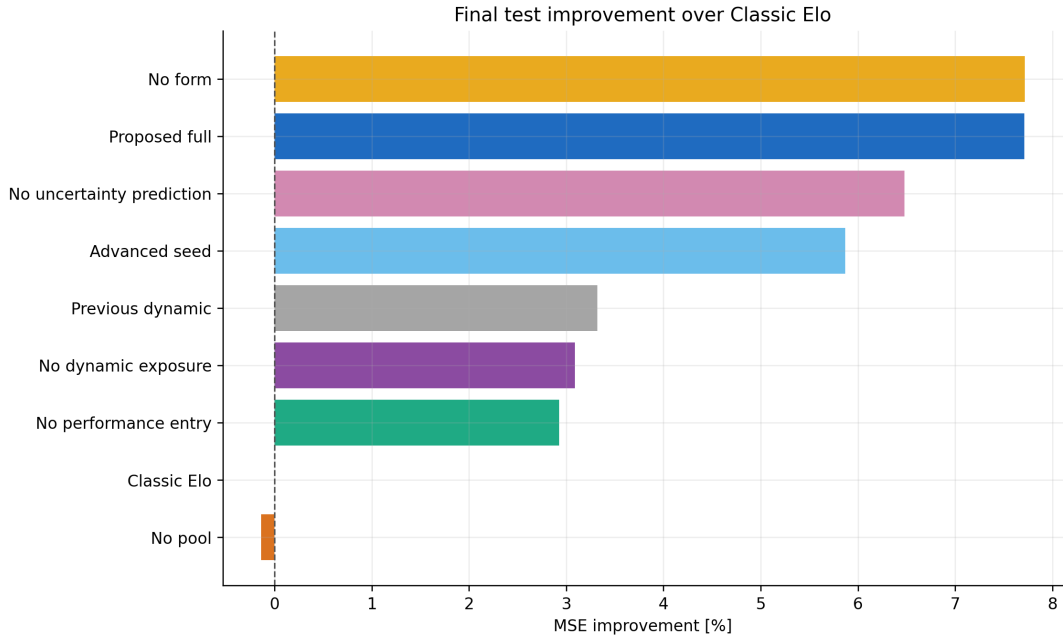


FIGURE 8.1: Test-period MSE improvement over the classical Elo baseline for selected variants. Positive values indicate lower MSE than classical Elo.

8.2 Train, validation and test behaviour

Table 8.2 reports representative metrics across train, validation and test periods. The proposed full model has similar MSE in all three scopes: 0.14372 on train, 0.14368 on validation and 0.14399 on test. The validation-test difference is small relative to the gap between the proposed model and the classical Elo baseline.

Model	Scope	Games	MSE	MAE
Classic Elo	train	6 675 441	0.158196	0.350628
Classic Elo	validation	3 268 828	0.156285	0.347702
Classic Elo	test	7 393 544	0.156026	0.347232
Advanced seed	train	6 675 441	0.147906	0.327952
Advanced seed	validation	3 268 828	0.147140	0.325939
Advanced seed	test	7 393 544	0.146868	0.325186
Proposed full	train	6 675 441	0.143722	0.322830
Proposed full	validation	3 268 828	0.143681	0.321576
Proposed full	test	7 393 544	0.143993	0.322149
No performance entry	train	6 675 441	0.151096	0.320806
No performance entry	validation	3 268 828	0.150541	0.320159
No performance entry	test	7 393 544	0.151462	0.322546
No pool	train	6 675 441	0.153461	0.329567
No pool	validation	3 268 828	0.153793	0.328491
No pool	test	7 393 544	0.156241	0.331646
No dynamic exposure	train	6 675 441	0.145733	0.323302
No dynamic exposure	validation	3 268 828	0.147845	0.323485
No dynamic exposure	test	7 393 544	0.151208	0.327758

TABLE 8.2: Train, validation and test metrics for representative variants.

Table 8.3 isolates the validation-test gap. The full model has a gap of about 0.00031. Variants without dynamic exposure or without pools have both worse absolute performance and larger

deterioration from validation to test. The result is consistent with the interpretation that pool-aware modelling improves future-period performance rather than only validation-period fit.

Model	Train MSE	Validation MSE	Test MSE	Test-Val	Improvement vs. Elo [%]
No form	0.143715	0.143674	0.143988	0.000314	7.72
Proposed full	0.143722	0.143681	0.143993	0.000312	7.71
Advanced seed	0.147906	0.147140	0.146868	-0.000272	5.87
No dynamic exposure	0.145733	0.147845	0.151208	0.003363	3.09
No performance entry	0.151096	0.150541	0.151462	0.000920	2.93
Classic Elo	0.158196	0.156285	0.156026	-0.000259	0.00
No pool	0.153461	0.153793	0.156241	0.002448	-0.14

TABLE 8.3: Generalization gap between training, validation and test periods. The last column reports MSE reduction relative to the classical Elo baseline on the final test period.

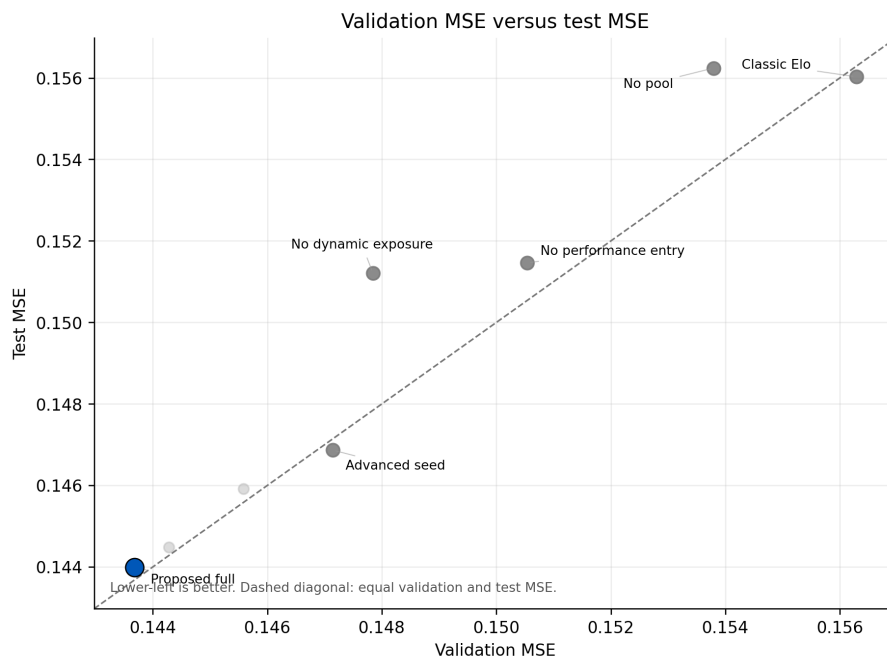


FIGURE 8.2: Validation MSE versus test MSE for selected variants. The dashed diagonal marks equal validation and test MSE; lower-left points are better.

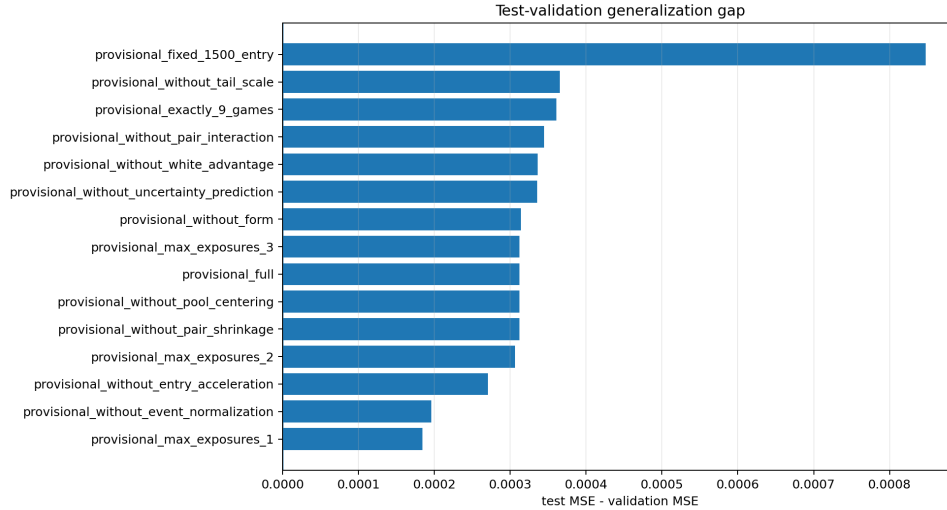


FIGURE 8.3: Test-validation generalization gap for selected variants.

The scatter plot and the gap table should be read together. The strongest variants cluster in a narrow MSE range, while the no-pool and no-dynamic-exposure variants remain clearly separated. The proposed model is close to the diagonal and retains its advantage on the final test period.

8.3 Ablation and component contribution

Table 8.4 summarises the component contribution analysis. The largest loss occurs when the pool component is removed. The full model improves over classical Elo by 7.71 percentage points, whereas the no-pool variant is slightly worse than the baseline. The lost improvement is therefore about 7.85 percentage points. This is the central empirical evidence for the pool-aware hypothesis.

Removed or changed component	Ablated MSE	MSE loss	Lost improvement pp
pool component	0.156241	0.012248	7.850
performance-based entry	0.151462	0.007469	4.787
dynamic exposure	0.151208	0.007215	4.624
non-1500 entry correction	0.146356	0.002363	1.514
uncertainty-aware prediction	0.145920	0.001927	1.235
white advantage	0.144538	0.000545	0.349
event normalization	0.144480	0.000487	0.312
tail-scale correction	0.144073	0.000081	0.052
pool-pair interaction	0.144050	0.000057	0.036
short-term form	0.143988	-0.000005	-0.003

TABLE 8.4: Component contribution estimated by ablation on the final test period.

The next two most important mechanisms are performance-based entry and dynamic exposure. Removing performance-based entry reduces the test improvement to approximately 2.93%, corresponding to a loss of about 4.79 percentage points. Removing dynamic exposure reduces the test improvement to approximately 3.09%, corresponding to a loss of about 4.62 percentage points. These results show that the pool component is most effective when players can enter the rating system with a more informative initial level and when their exposure to pools can change over time.

Uncertainty-aware prediction has a smaller but still visible effect. Removing this component reduces the test improvement from approximately 7.71% to 6.48%. Removing white advantage

reduces the improvement to approximately 7.36%. These components do not explain the main gain, but they improve the expected-score model.

Some components have little practical effect in the final configuration. The variant without short-term form is almost indistinguishable from the full model and is marginally better on the final test period. Pairwise pool interactions also add little compared with the main pool component: the no-pair-interaction variant still improves over the classical baseline by approximately 7.68%. These results motivate the simplified version discussed later.

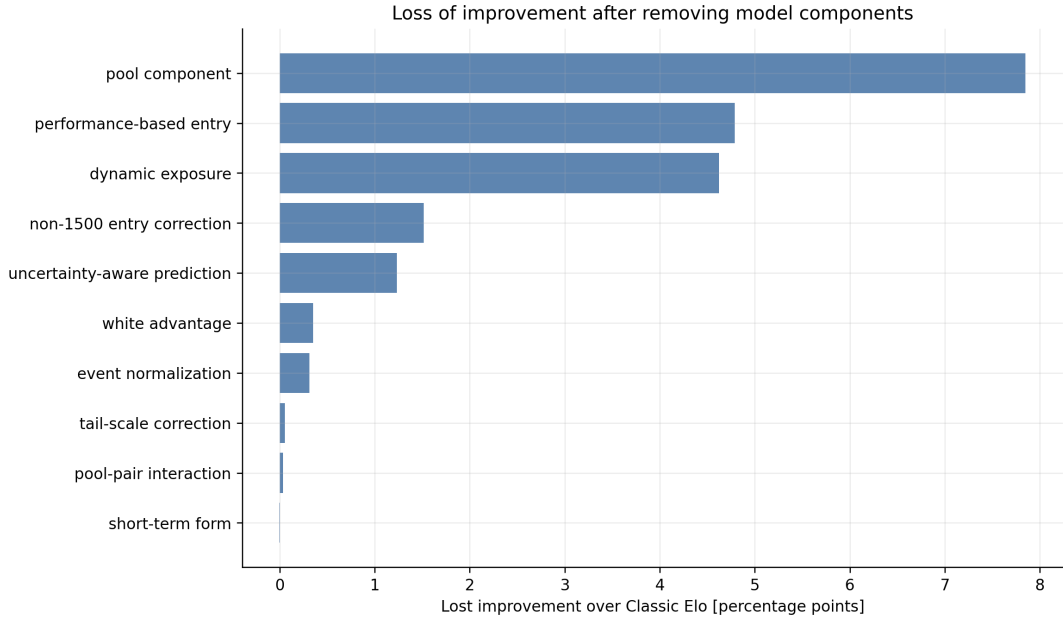


FIGURE 8.4: Lost improvement after removing or changing selected model components. Larger bars indicate components with stronger predictive contribution.

8.4 Monthly stability

Aggregate MSE can conceal temporal instability. Table 8.5 reports monthly improvement statistics. The proposed full model has mean monthly improvement 7.62%, median improvement 7.81%, and positive improvement in 86.85% of months. The worst month still shows positive improvement relative to classical Elo.

Model	Mean [%]	Median [%]	Worst month [%]	Positive months [%]
No form	7.62	7.82	1.58	86.9
Proposed full	7.62	7.81	1.58	86.9
Advanced seed	5.70	5.76	0.86	86.9
No dynamic exposure	5.66	5.97	1.58	86.9
No performance entry	3.82	3.98	0.46	86.9
Previous dynamic	3.28	3.29	1.21	86.9
No pool	2.33	2.72	-2.13	67.6

TABLE 8.5: Monthly stability of improvement over classical Elo.

Figures 8.5 and 8.6 show monthly behaviour from 2015 onward. The earlier period is used for historical replay and pool construction; starting the plot at 2015 keeps the visual focus on the parameter-selection and evaluation horizon. The vertical dashed lines mark the beginning of

validation and test periods. The no-pool variant is consistently weaker than the full model, while the no-form variant remains almost identical to it.

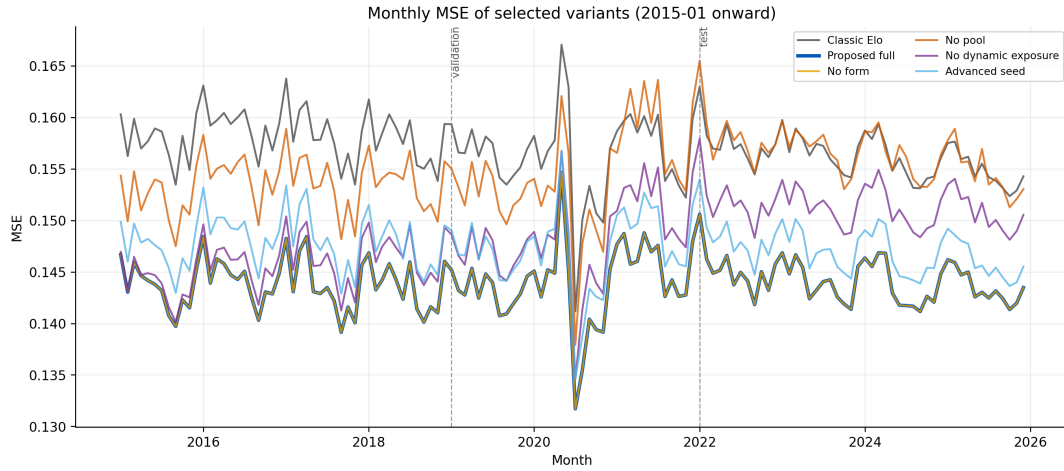


FIGURE 8.5: Monthly MSE for selected model variants. Lower values are better. The dashed vertical lines mark validation and test starts.

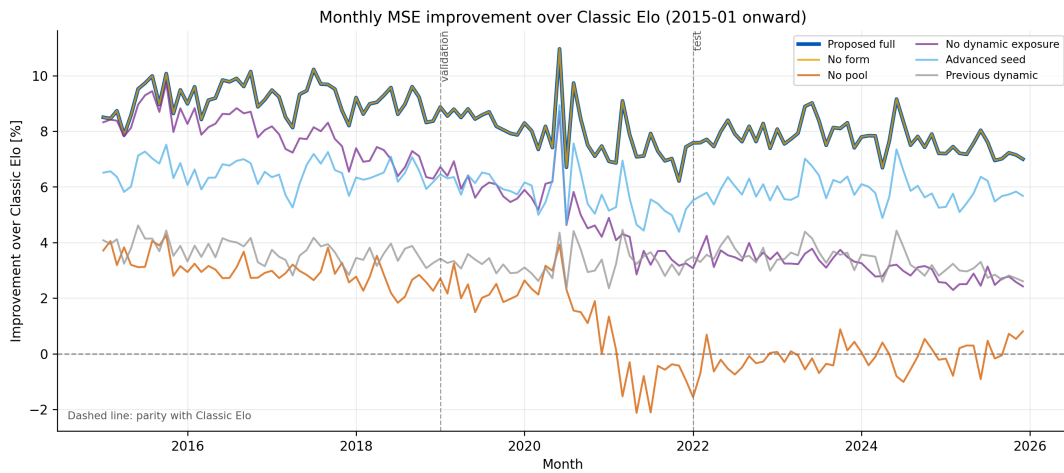


FIGURE 8.6: Monthly MSE improvement over classical Elo for selected variants. The horizontal dashed line denotes parity with classical Elo, not a separate model.

8.5 Comparison with official FIDE ratings

On the shared subset of the test period, official FIDE ratings have MSE 0.149907, while the proposed full model has shared-subset MSE 0.139329. This comparison is not a claim that the experimental system is ready to replace the official rating list. Official ratings also serve governance, transparency, stability, title regulation and tournament-administration purposes. The shared-subset result only shows that the experimental model extracts predictive information not fully captured by the one-dimensional official rating value.

The difference is consistent with the model design. Official ratings are public scalar values. The proposed expected score depends on base rating, uncertainty, colour, pool exposure and pool context. The additional context improves prediction in the historical test period, while Chapter 10 discusses the practical constraints of turning such an internal model into a public rating list.

Chapter 9

Robustness and Sensitivity Experiments

9.1 Expected-score curves

The proposed model modifies not only the rating update, but also the mapping from rating difference to expected score. Figure 9.1 shows expected-score curves at different rating levels. A 400-point advantage has expected score 0.909 in classical Elo. In the proposed model the corresponding expected score can be sharper or flatter depending on average level and uncertainty. Table 9.1 gives selected values.

Curve	Expected score at +400	Scale
Classic Elo	0.909	
Final, level 1500	0.931	353.1
Final, level 2000	0.873	477.5
Final, level 2500	0.822	601.9
Final, level 1500, unc. 1	0.796	677.4

TABLE 9.1: Example expected-score values for a 400 point rating advantage.

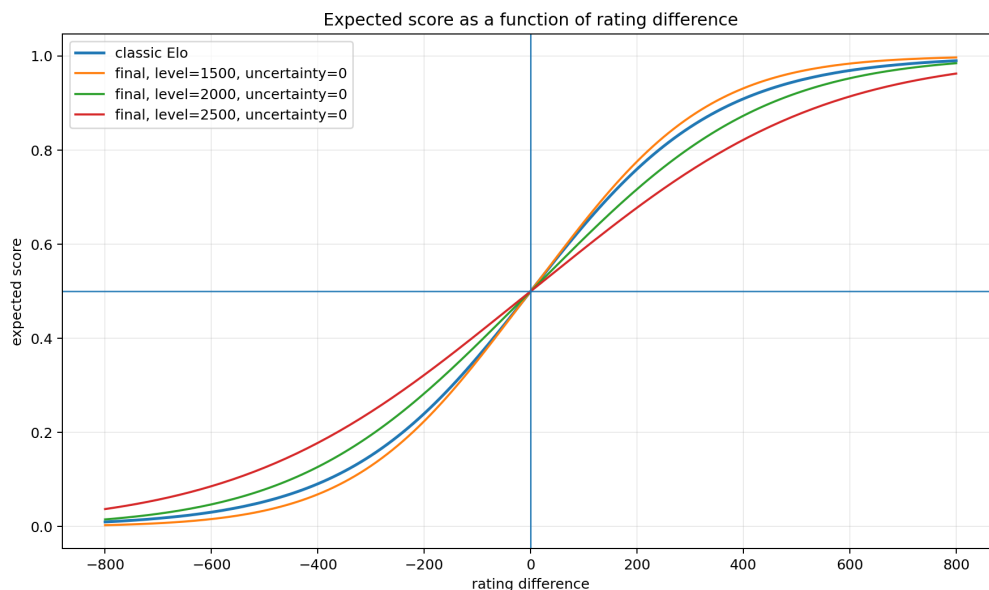


FIGURE 9.1: Expected-score curves for classical Elo and the proposed model at different average rating levels.

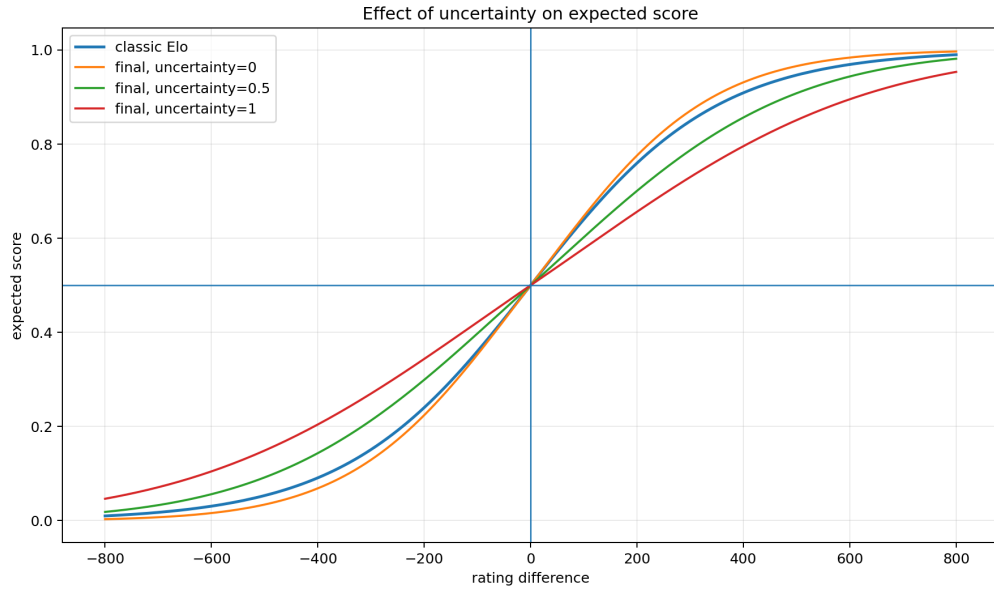


FIGURE 9.2: Effect of uncertainty on the expected-score curve. Higher uncertainty flattens predictions.

At higher levels, draws are more frequent and a nominal rating difference does not necessarily translate into the same expected score as in lower-level games. Uncertainty also supports less extreme predictions. The calibration experiment below checks whether these expected scores remain empirically aligned with observed results.

9.2 Sensitivity to pool strength

Table 9.2 and Figure 9.4 show what happens when the learned pool weight is multiplied by different factors. Removing the pool contribution almost eliminates the advantage on the test period. Increasing the multiplier from 0 to 1 steadily improves performance. Values above 1 are affected by clipping to the allowed parameter range and should not be interpreted as unrestricted scaling.

Pool-weight multiplier	Test MSE	Improvement vs. Elo [%]
0.00	0.156158	-0.08
0.25	0.155365	0.42
0.50	0.149664	4.08
0.75	0.145553	6.71
1.00	0.143993	7.71
1.25	0.143993	7.71
1.50	0.143993	7.71
2.00	0.143993	7.71

TABLE 9.2: Sensitivity to the pool-weight multiplier. Improvements are test-period MSE reductions relative to the classical Elo baseline. Values above 1 are affected by clipping to the allowed parameter range.

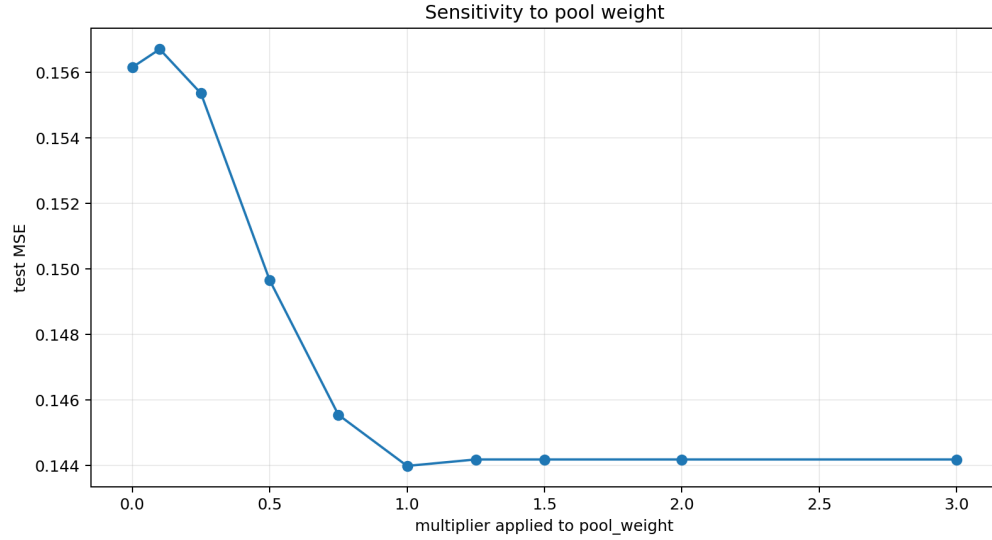


FIGURE 9.3: Test MSE as a function of the pool-weight multiplier.

The absolute MSE decreases sharply when the multiplier moves from 0.25 to 0.75 and reaches its best region around the validation-selected value. The curve indicates a broad interval in which a substantial pool correction is better than removing the correction.

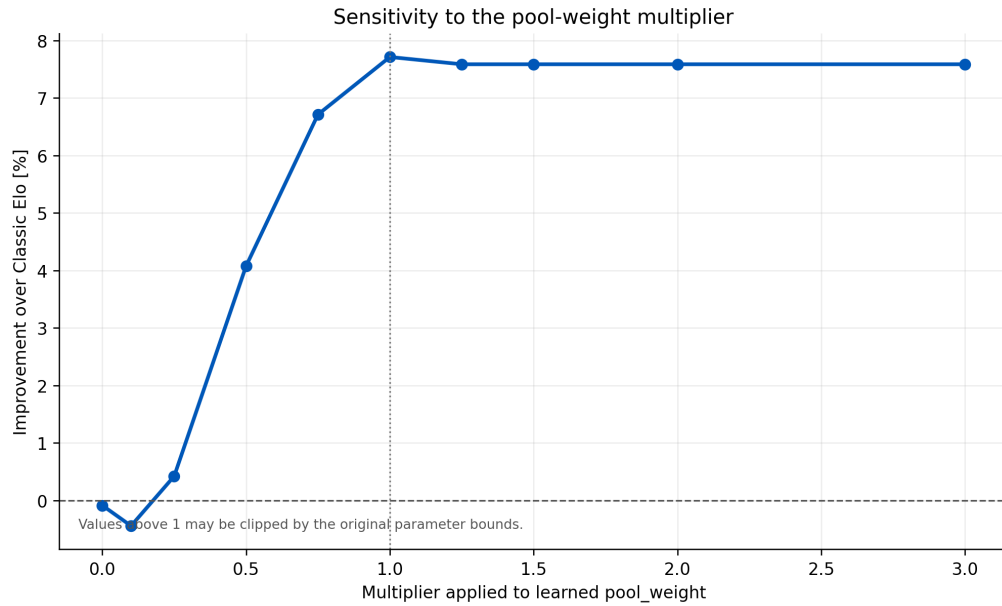


FIGURE 9.4: Sensitivity of test improvement to the pool-weight multiplier.

The experiment identifies the pool component as the main predictive mechanism beyond classical Elo. Without it, the remaining components do not preserve the same advantage.

9.3 Sensitivity of provisional entry

The provisional-entry mechanism is analysed in Table 9.3. Increasing the target number of provisional games from 3 to 12 improves performance. A target of 15 is marginally better on the

test period, but the model was selected on validation and uses 12. The relevant pattern is the stability of the region around 9–15 games.

Parameter	Value	Test MSE	Improvement vs. Elo [%]
target games	3.00	0.145079	7.02
target games	6.00	0.144473	7.40
target games	9.00	0.144134	7.62
target games	12.00	0.143993	7.71
target games	15.00	0.143962	7.73
blend	0.00	0.151462	2.93
blend	0.25	0.146389	6.18
blend	0.50	0.144504	7.39
blend	0.72	0.143995	7.71
blend	1.00	0.144170	7.60
K multiplier	1.00	0.144111	7.64
K multiplier	1.50	0.143968	7.73
K multiplier	1.80	0.143992	7.71
K multiplier	2.50	0.144351	7.48
K multiplier	3.50	0.145573	6.70

TABLE 9.3: Sensitivity of the provisional-entry mechanism. Improvements are test-period MSE reductions relative to the classical Elo baseline.

The blend parameter is particularly informative. If the blend is zero, the performance-based part is removed and improvement drops to about 2.93%. Moderate blending around 0.7 is best. Full trust in the early performance estimate is slightly worse than the validation-selected value, which indicates that early performance is informative but noisy.

The provisional K multiplier also has a stable optimum. A multiplier around 1.5–1.8 works well. Too large a multiplier makes new-player ratings excessively volatile and reduces test performance.

Figures 9.5, 9.6 and 9.7 present the same analysis visually. The target-games curve is relatively flat after about nine games, so the exact boundary is not fragile. The blend curve shows a clear improvement when early performance is used, followed by a mild decline when the blend is pushed to one. The K -multiplier curve confirms that acceleration is beneficial only in moderation.

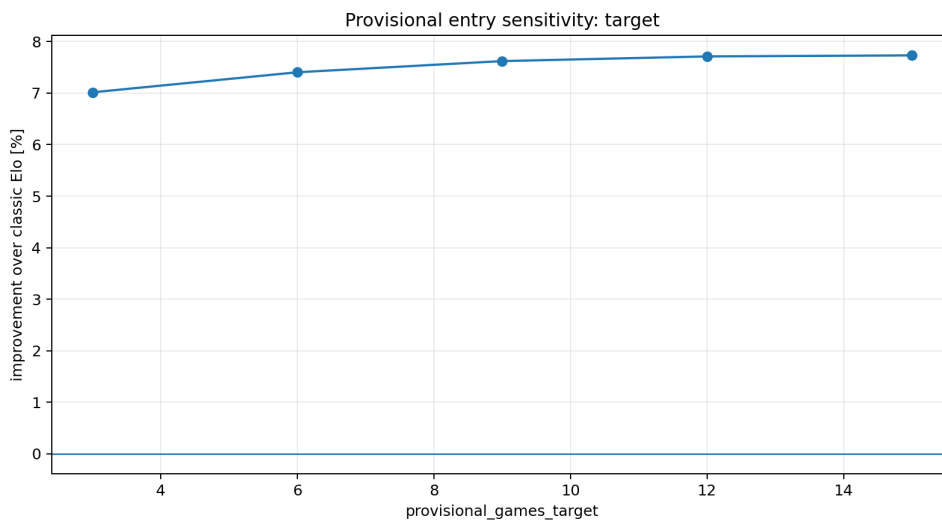


FIGURE 9.5: Sensitivity to the number of provisional games.

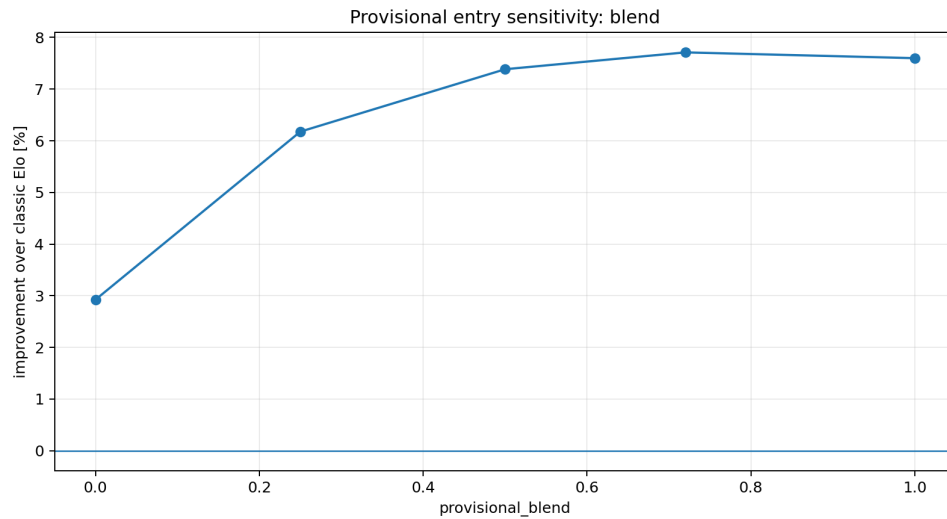


FIGURE 9.6: Sensitivity to the provisional performance blend.

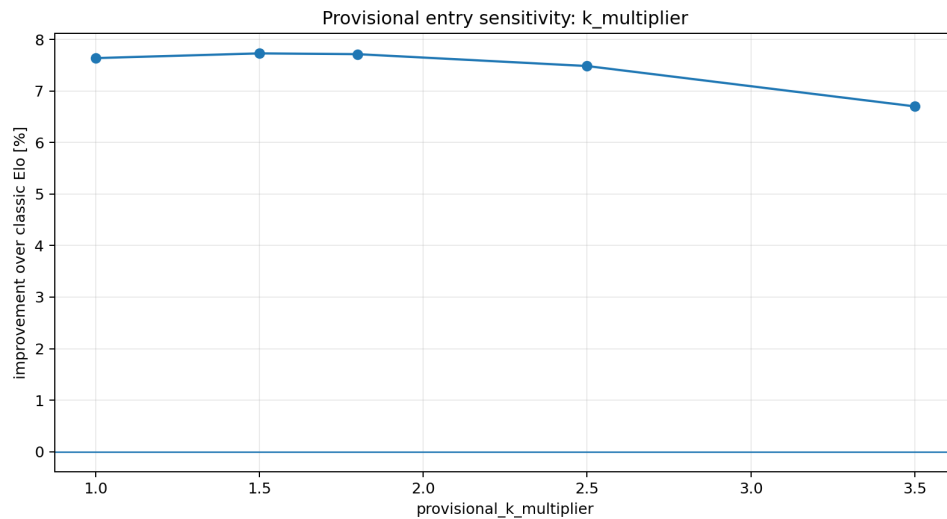


FIGURE 9.7: Sensitivity to the provisional update multiplier.

9.4 Calibration of expected scores

Calibration is evaluated by grouping test predictions into probability buckets. The weighted mean absolute calibration error is 0.007481 over 7,393,544 test games. The average absolute difference between predicted and observed expected score is therefore below one percentage point. Table 9.4 reports aggregated buckets.

Pred. range	Games	Predicted	Observed	Obs.-Pred.
0.00–0.10	145 534	0.066	0.080	0.013
0.10–0.20	440 390	0.156	0.166	0.010
0.20–0.30	713 067	0.253	0.263	0.010
0.30–0.40	900 981	0.351	0.361	0.009
0.40–0.50	1 027 135	0.451	0.462	0.010
0.50–0.60	1 117 169	0.550	0.560	0.010
0.60–0.70	1 128 238	0.650	0.656	0.007
0.70–0.80	993 166	0.748	0.750	0.002
0.80–0.90	678 907	0.845	0.843	-0.002
0.90–1.00	248 957	0.934	0.929	-0.005

TABLE 9.4: Calibration of expected-score predictions on the test set.

The calibration plot in Figure 9.8 is generally close to the diagonal. There is slight overconfidence in the most extreme probability regions: very low predicted scores are a little too low, and very high predicted scores are a little too high. This pattern is consistent with calibration literature [51, 54, 55, 56].

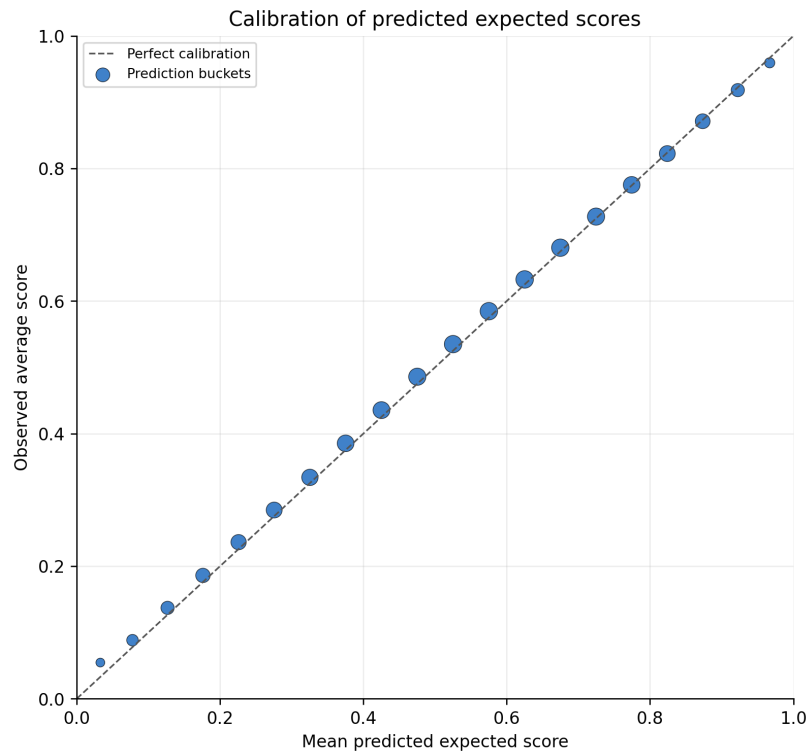


FIGURE 9.8: Calibration of predicted expected scores on the final test period.

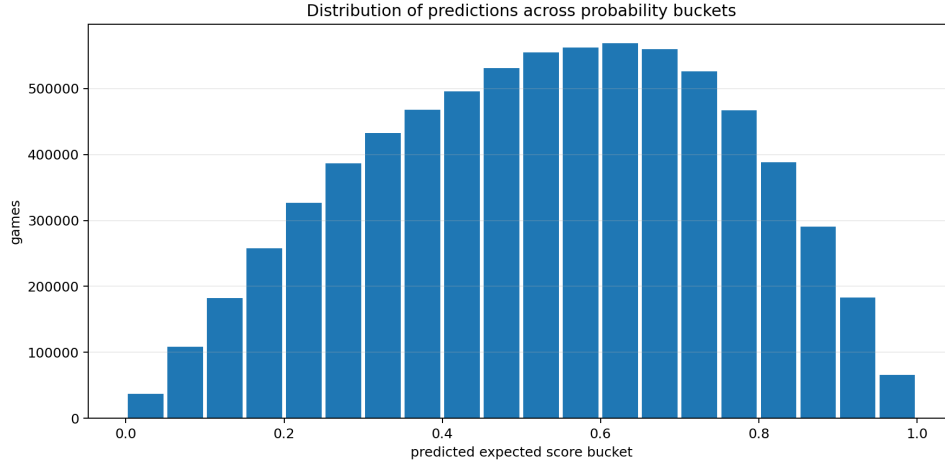


FIGURE 9.9: Distribution of test games across expected-score buckets.

The bucket-count plot gives the scale of each calibration region. Extreme probability ranges contain fewer games than central ranges, so the tails of the calibration curve should not be overinterpreted. The central buckets dominate the weighted calibration error, which is why mild overconfidence in the tails does not contradict the low overall calibration error.

9.5 Artificial global rating shocks

A stress test artificially shifts internal ratings at the start of the test period and observes the excess monthly MSE. Table 9.5 reports global shocks of ± 150 and ± 300 points. Positive shocks recover more quickly than negative shocks. A $+150$ shock first falls below the threshold in April 2022. A $+300$ shock first falls below the threshold in April 2023. Negative shocks leave a longer residual effect.

Shock	Max excess MSE	Mean excess MSE	First below threshold
-300	0.007906	0.005171	not reached
-150	0.002352	0.001548	not reached
150	0.000722	0.000155	2022-04
300	0.002916	0.001250	2023-04

TABLE 9.5: Global artificial rating shock experiment.

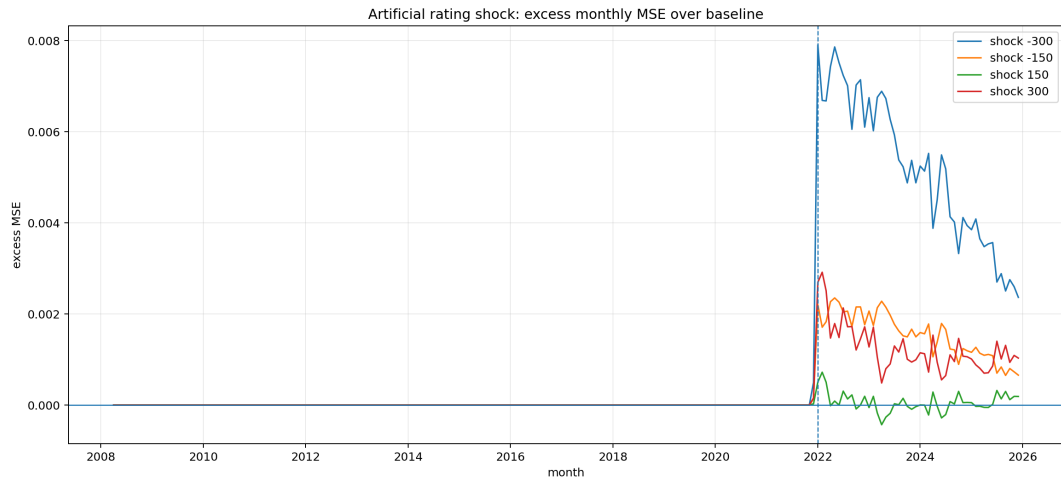


FIGURE 9.10: Excess monthly MSE after artificial global rating shocks.

The experiment is a stress test rather than a simulation of a specific regulation change. It shows that the rating system can partially correct a large miscalibration, but global shifts are difficult and can leave long-term effects.

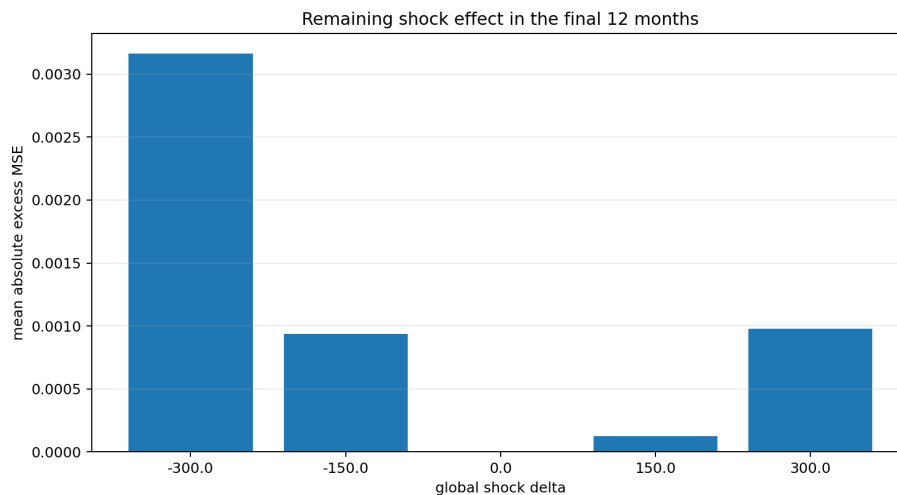


FIGURE 9.11: Remaining global-shock effect in the last twelve months of the test period.

The remaining-effect plot confirms the asymmetry between positive and negative shocks. Negative shifts are harder to eliminate because many players must climb back through future wins, while losses and draws against the shifted population can keep the system compressed. Positive shifts, especially moderate ones, are corrected more quickly. The asymmetry indicates that rating-scale interventions can have path-dependent consequences.

9.6 Pool-specific shock recovery

The pool-specific version of the shock experiment is more directly related to the pool-aware mechanism. Instead of shifting all players, it shifts players from one latent pool. Table 9.6 shows results for the five largest non-default pools. The excess MSE is much smaller than in the global-

shock experiment, which indicates that local pool-level miscalibration is easier for the model to absorb than a global rating-scale shift.

Pool	Shock	Mean abs excess	Final 12m
1	-300	0.000426	0.000192
1	-150	0.000124	0.000071
1	150	0.000032	0.000023
1	300	0.000156	0.000034
2	-300	0.000180	0.000071
2	-150	0.000031	0.000015
2	150	0.000135	0.000067
2	300	0.000417	0.000204
3	-300	0.000206	0.000074
3	-150	0.000044	0.000018
3	150	0.000090	0.000035
3	300	0.000303	0.000118
4	-300	0.000215	0.000150
4	-150	0.000061	0.000066
4	150	0.000049	0.000021
4	300	0.000174	0.000046
5	-300	0.000184	0.000056
5	-150	0.000047	0.000015
5	150	0.000040	0.000017
5	300	0.000164	0.000066

TABLE 9.6: Pool-specific artificial rating shock experiments for the five largest non-default pools.

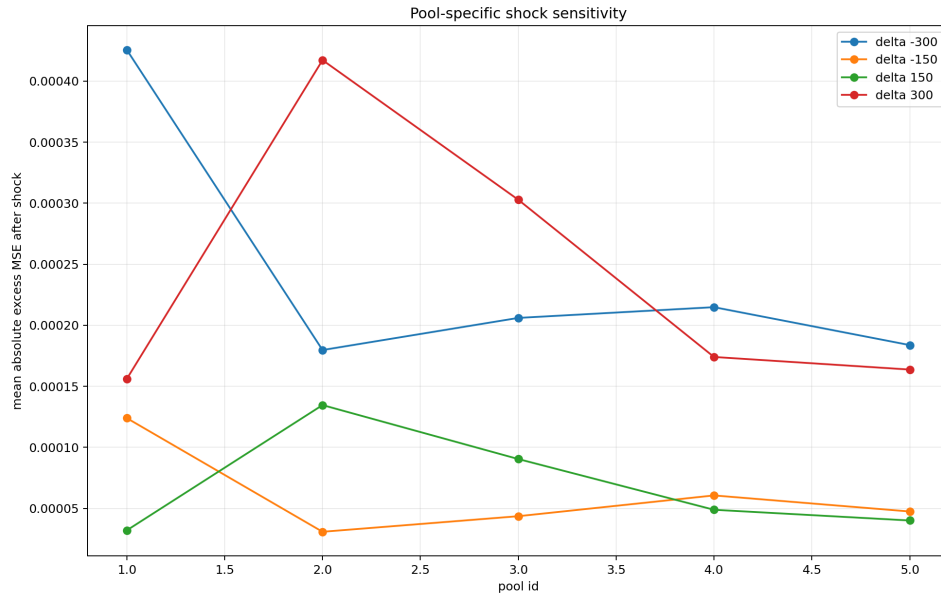


FIGURE 9.12: Mean excess MSE after pool-specific artificial shocks.

The result supports the interpretation that pools localise systematic errors. When one environment is shifted relative to the rest of the world, a pool-aware system has a mechanism to correct that shift without distorting every player's rating equally.

9.7 Sensitivity to update speed and scale

The model is sensitive to rating-update speed and expected-score scale, as shown in Table 9.7. Too small a base K slows adaptation. Too large a value reaches the allowed parameter range and does not improve the model. The expected-score scale has a clear optimum: both too sharp and too flat curves worsen test MSE.

Parameter	Multiplier	Test MSE	Improvement vs. Elo [%]
base K	0.50	0.147008	5.78
base K	0.75	0.144480	7.40
base K	1.00	0.143993	7.71
base K	1.25	0.144199	7.58
base K	1.50	0.144199	7.58
base K	2.00	0.144199	7.58
scale base	0.50	0.145916	6.48
scale base	0.75	0.144777	7.21
scale base	1.00	0.143993	7.71
scale base	1.25	0.144808	7.19
scale base	1.50	0.146477	6.12
scale base	2.00	0.147632	5.38

TABLE 9.7: Sensitivity to the base update coefficient and expected-score scale. Improvements are test-period MSE reductions relative to the classical Elo baseline.

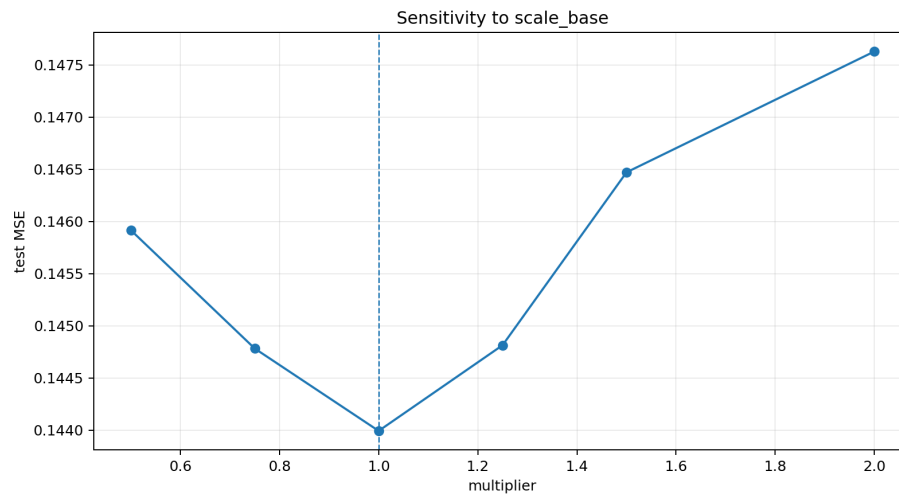


FIGURE 9.13: Sensitivity to the base expected-score scale.

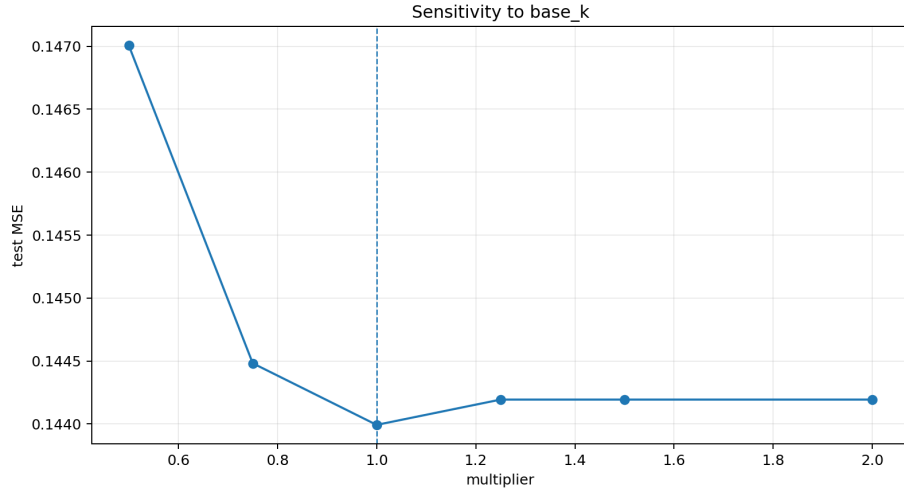


FIGURE 9.14: Sensitivity to the base update coefficient.

The K sensitivity experiment complements the provisional-entry analysis. A very low base update speed makes the model slow to react even if other components are present. Simply making the whole system more aggressive does not solve the problem. The best region is close to the validation-selected value, while large multipliers are limited by the allowed parameter range. The improvement therefore comes from the structure of the model rather than from a uniformly faster update rule.

9.8 White advantage

The model learns a white advantage of about 36 rating points. Table 9.8 and Figure 9.15 show that values around 30–36 are best. Setting white advantage to zero is worse, but the loss is modest compared with removing pools or provisional entry. Colour is therefore a useful correction, not a core explanation of the main improvement.

White advantage	Test MSE	Improvement vs. Elo [%]
0.0	0.144752	7.23
10.0	0.144360	7.48
20.0	0.144107	7.64
30.0	0.143993	7.71
36.1	0.143993	7.71
45.0	0.144084	7.65
60.0	0.144486	7.40
70.0	0.144923	7.12

TABLE 9.8: Sensitivity to the white-advantage correction. Improvements are test-period MSE reductions relative to the classical Elo baseline.

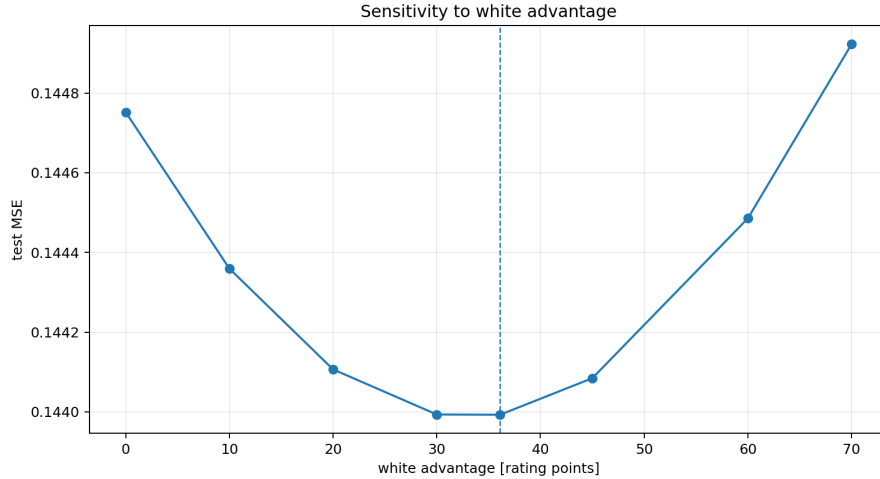


FIGURE 9.15: Sensitivity to the white-advantage correction.

9.9 Rating-scale diagnostics

The proposed system improves prediction, but its raw effective rating scale differs from classical Elo. Table 9.9 shows that the full model has a higher median effective rating and a wider p99–p01 range than the baseline. Figure 9.16 visualises the monthly range.

The expected-score curves from Section 9.1 provide context for this diagnostic. The model changes not only player coordinates on the rating scale, but also the mapping from rating difference to expected score. A wider effective-rating distribution therefore does not automatically imply more extreme predictions. The evaluated quantity is the final expected score after effective rating, scale, uncertainty and pool context have been combined.

Model	Mean median	Mean p99-p01	Mean p90-p10
Classic Elo	1485.2	539.2	209.8
Proposed full	1701.0	1213.9	671.7
No form	1701.2	1213.0	671.4
No pool	1384.6	1020.5	530.8

TABLE 9.9: Monthly rating-scale diagnostics across 185 months.

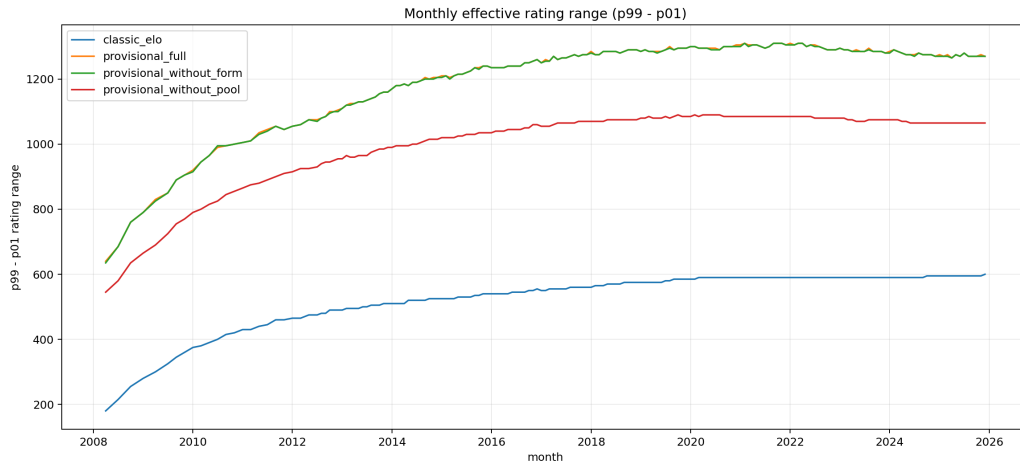


FIGURE 9.16: Monthly effective rating range measured by the p99–p01 width.

This is not necessarily a predictive flaw. The model optimises expected-score accuracy, not visual compatibility with the traditional FIDE scale. It does, however, matter for presentation. A public version would need a display calibration or normalisation step so that reported ratings remain understandable and comparable.

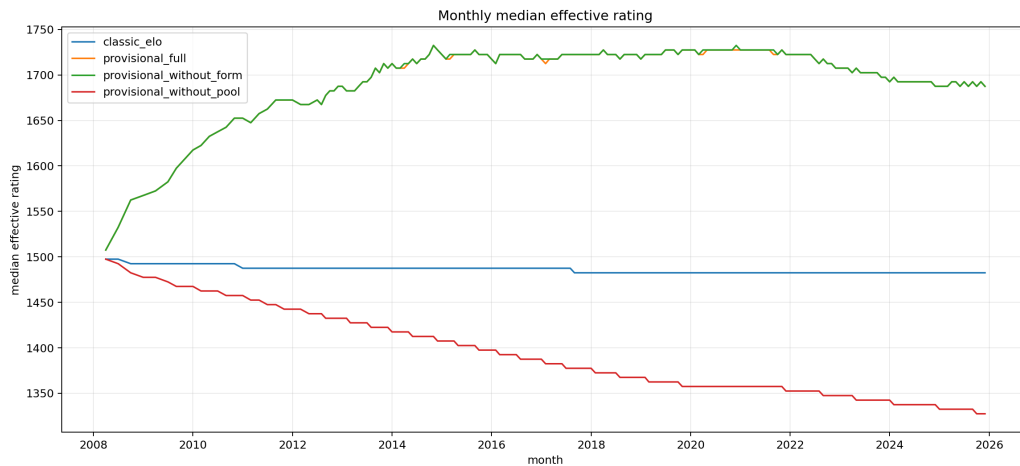


FIGURE 9.17: Monthly median effective rating for selected model variants.

The median-rating plot shows that the proposed model moves to a different internal scale during replay. This does not affect expected-score evaluation because predictions depend on effective differences and scale parameters. It affects only the interpretation of raw rating values. A user-facing rating list should therefore not expose raw effective ratings without anchoring them to a familiar scale.

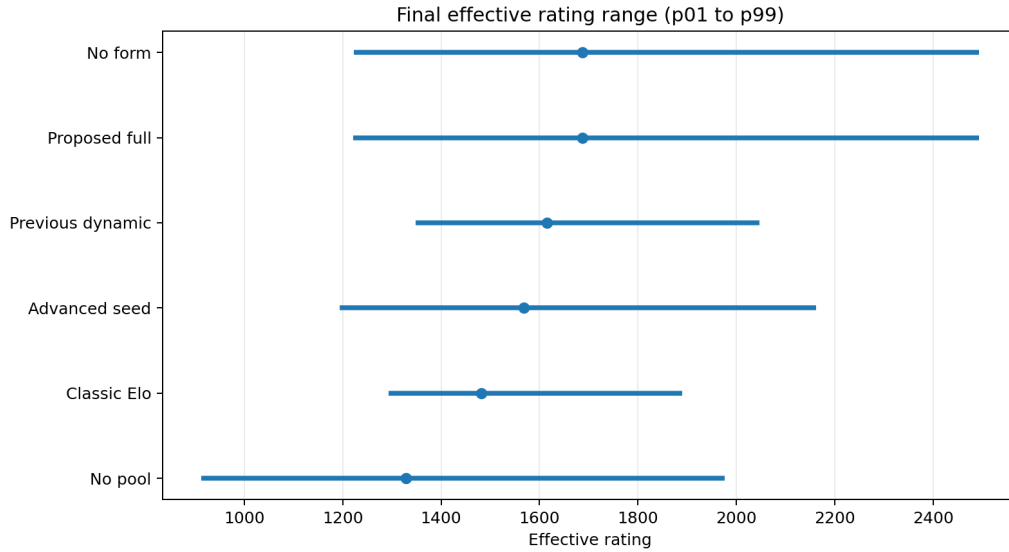


FIGURE 9.18: Final effective rating ranges by model.

The final-range diagnostic is consistent with the monthly scale analysis. The full model uses a much wider effective range than classical Elo. This wider range should not be read directly as a claim that the best players are publicly “worth” that many points. It is an internal coordinate system that combines base rating and contextual corrections. The practical display problem is therefore separate from the predictive problem.

9.10 Model simplification

The model simplification report in Table 9.10 shows that several components can be removed with minimal loss. The variant without short-term form is marginally better on the test period. Removing pair shrinkage or pool centering has almost no effect. Reducing maximum exposures from 3 to 2 causes only a small loss. Removing pair interactions is also relatively safe.

Model	Test MSE	Improvement vs. Elo [%]	Gap vs. full [pp]
No form	0.143988	7.72	-0.003
No pair shrinkage	0.143992	7.71	-0.000
Proposed full	0.143993	7.71	0.000
Max exposures 3	0.143993	7.71	0.000
No pool centering	0.143993	7.71	0.000
Max exposures 2	0.144028	7.69	0.023
No pair interaction	0.144050	7.68	0.036
No tail scale	0.144073	7.66	0.052
No entry acceleration	0.144111	7.64	0.076
Exactly 9 provisional games	0.144134	7.62	0.091
Max exposures 1	0.144381	7.46	0.249
No event normalization	0.144480	7.40	0.312

TABLE 9.10: Model variants closest to the proposed full system. The gap is measured in percentage points of improvement relative to the full model.

This suggests a practical two-level interpretation. The full model is useful for experimentation and ablation. A simplified model for communication could keep base rating, uncertainty, performance-based entry, dynamic pool exposure and white advantage, while removing short-term

form and possibly pair interactions. Such a simplification would preserve most predictive benefit and improve interpretability.

Chapter 10

Discussion

10.1 Predictive role of latent pools

The experimental results indicate that the chess player population is not fully homogeneous from the perspective of expected-score prediction. A classical Elo-style baseline treats the rating scale as global and uniform. The pool-aware model keeps the sequential structure of Elo, but adds state variables describing the interaction environment in which a player has been observed.

The strongest evidence comes from the ablation experiments. Removing the pool component eliminates almost the entire predictive advantage over the classical baseline. Removing dynamic exposure or performance-based provisional entry also produces a substantial loss. The improvement is therefore not caused by a single additional parameter or by a small calibration adjustment. It comes mainly from modelling two related effects: partially separated player environments and the way players enter or move between them.

The comparison with official FIDE ratings on the shared subset leads to the same general conclusion. The model is not only better than an internal Elo baseline, but also improves expected-score prediction relative to the available official-rating comparison. This should be interpreted as a predictive result, not as a claim that the model is ready to replace an official rating list.

10.2 Interpretation of latent pools

The descriptive analysis of pools gives a coherent picture of the discovered structure. Several pools are federation-dominated, others correspond to regional playing environments, and one major pool is associated with globally connected elite tournaments. This is consistent with the way chess is organised: many players compete mainly inside national or regional circuits, while stronger and more active players create bridges between those circuits.

The pool labels are nevertheless internal modelling objects. They are inferred from games, not assigned from geography or administrative categories. A pool should therefore not be read as a direct statement about a federation, region or group of players. Pool 0 is especially different from the others, since it functions as a default or weak-assignment pool and contains most players.

The learned pool offsets require the same caution. They are predictive corrections inside a fitted replay model. They are not public estimates of national strength, nor are they suitable as direct rating differences between federations. The rating-scale diagnostics show that raw effective ratings require additional calibration before they could be displayed as a public rating list.

10.3 Provisional players and changing populations

The junior and age-profile analyses provide an important explanation for the performance of the provisional mechanism. The recent FIDE population contains a large and increasing share of young players. Many of them improve quickly and are poorly represented by a fixed initial rating. A system that enters every new player with the same value adapts slowly and can produce distorted predictions during the early part of a player’s rating history.

Performance-based provisional entry addresses this problem directly. It estimates a more informative initial level from early results, while uncertainty and higher early update speed allow the rating to stabilise over time. Dynamic pool exposure then connects a player to the environments in which their games actually occur, instead of assuming that every new player immediately belongs to the same global context.

10.4 Full model and practical simplification

The full model is useful as an experimental system because it allows the contribution of many mechanisms to be tested. The ablation study, however, suggests that some components are not necessary for most of the gain. Short-term form has negligible effect, and pairwise pool interactions add little compared with the main pool component. Pool centering and pairwise shrinkage also have little practical influence in the final configuration.

A more practical variant would retain the base rating, uncertainty, performance-based provisional entry, dynamic pool exposure, colour correction and event-level normalisation. Such a model would be easier to explain and monitor while preserving the central empirical result: graph-derived interaction communities improve sequential expected-score prediction.

10.5 Interpretation and deployment constraints

The main deployment issue is not predictive validity, but interpretation of the rating scale. The effective ratings produced by the model have a wider distribution and a higher median than the classical Elo baseline. This means that the numerical values should not be read as direct replacements for current FIDE Elo points. A player rated 2400 in one system and 2400 in another system would not necessarily occupy the same position in the underlying distribution.

This does not prevent the model from being used as a rating system. It means that a public version would need a clearly defined display scale. One option would be to publish the model on its own scale, similarly to how different platforms already use different rating systems. Another option would be to transform effective ratings onto a calibrated public scale, for example by anchoring selected percentiles or reference groups. In both cases, the main requirement is clarity: users must know whether a displayed number is intended to be comparable with FIDE Elo or whether it belongs to a new scale.

A related issue is the interpretation of pool corrections. Pool offsets improve expected-score prediction, but they are internal components of the model. They should not be presented as public statements about the strength of federations, regions or demographic groups. In a deployed system, these corrections would be better exposed through the final rating and uncertainty values rather than as standalone labels.

10.6 Practical implications

A public rating system affects tournament invitations, title paths, pairings and player reputation. Predictive accuracy is therefore only one requirement. Transparency, resistance to manipulation, scale stability and acceptance by players are also necessary. Pool-aware corrections would require careful documentation and monitoring, especially because pool offsets can be misunderstood as statements about federations or regions.

The experiments are better read as evidence for a modelling direction than as a final public-rating proposal. Interaction structure contains useful information, and a rating system can use that information while remaining sequential. Turning this into an official system would require a simpler parameterisation, calibrated display ratings and clear rules for updating latent pools over time.

Chapter 11

Conclusion

A dynamic pool-aware rating model was developed and evaluated on historical FIDE standard-rating data. The model extends an Elo-style sequential replay with uncertainty, performance-based provisional entry, latent pools inferred from the player interaction graph and dynamic pool exposure.

The validation-selected model improves test MSE by about 7.7% relative to the classical Elo baseline. On the subset where an official FIDE comparison is available, it also improves expected-score prediction relative to the official-rating baseline. The improvement is stable across the test period and is not explained by a single exceptional month.

The ablation study identifies the main mechanisms behind the result. The pool component is essential. Dynamic exposure and performance-based provisional entry are also central. Uncertainty, colour correction and event-level normalisation provide additional support. Short-term form and pairwise pool interactions are not important enough to justify the extra complexity in a practical simplified version.

The descriptive analysis shows that the learned pools correspond to real chess environments. Some are federation-dominated, others are regional, and one major pool represents a globally connected elite tournament environment. Junior analysis also shows a strong increase in under-21 participation over time, which helps explain why fixed provisional entry is insufficient for recent FIDE data.

The learned pool offsets are internal predictive quantities. They should not be presented as standalone judgements about federations, regions or groups of players. A public version of the system would need a clear display policy: either a separate rating scale with its own interpretation, or a calibrated transformation onto a public-facing scale. The important requirement is that users understand whether the displayed number is intended to be comparable with current FIDE Elo or belongs to a different model.

The results suggest that chess ratings can benefit from modelling the structure of the player population, not only individual player strength. A sequential rating system can remain close to Elo in operation while using graph-derived context to handle partially separated playing environments. Further work could focus on the public display scale and a simpler version of the model more suitable for transparent use.

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Appendix A

Emergent Structure of Latent Player Pools

A.1 Scope of the analysis

The main experiments indicate that latent pools have a large predictive effect. The pool identifiers themselves are numerical labels, so their interpretation requires a separate descriptive analysis. This chapter examines the composition of the discovered pools using player metadata: federation, birth year, rating level, top-player status and cross-pool game volume.

Pool 0 is treated separately throughout the analysis. It contains 465,273 of 597,530 players, or about 77.9% of all players. It is a default or weak-assignment pool rather than an ordinary interaction community. The descriptive analysis therefore focuses mainly on non-zero pools.

A.2 Federation composition

Table A.1 reports the largest non-default pools and their dominant federations. Several pools are strongly aligned with individual federations. Pool 8 is dominated by Italy, pool 2 by France, pool 3 by Spain, pool 7 by Poland and pool 4 by India. Other pools have a regional structure: pool 6 contains Brazil, Argentina and Peru; pool 10 contains Denmark, Sweden and Norway; pool 12 contains Turkey, Georgia, Armenia and Azerbaijan.

Pool	Players	Top share [%]	Fed. count	Fed. 1	Fed. 2	Fed. 3	Fed. 4	Fed. 5
1	12 045	68.1	67	GER (8 203)	NED (1 399)	BEL (1 212)	UNK (869)	FRA (37)
2	10 719	90.9	69	FRA (9 741)	UNK (344)	LUX (132)	GER (112)	BEL (67)
3	8 917	90.7	64	ESP (8 087)	UNK (545)	AND (43)	GER (27)	PER (18)
4	7 391	81.3	38	IND (6 012)	UNK (951)	BAN (335)	VIE (12)	USA (10)
5	6 652	55.4	36	CZE (3 684)	SVK (1 981)	UNK (700)	GER (119)	POL (33)
6	4 938	35.1	34	BRA (1 734)	ARG (1 239)	PER (733)	UNK (533)	BOL (221)
7	4 910	88.9	32	POL (4 364)	UNK (398)	GER (32)	CAN (29)	UKR (15)
8	4 794	92.2	45	ITA (4 421)	UNK (211)	GER (25)	SUI (25)	FRA (16)
9	4 475	79.7	54	HUN (3 566)	UNK (556)	JPN (66)	GER (46)	AUT (29)
10	4 177	44.8	47	DEN (1 873)	SWE (874)	NOR (814)	UNK (455)	GER (28)
11	3 845	78.5	59	RUS (3 020)	UNK (383)	FID (75)	GER (55)	UKR (43)
12	3 781	45.9	62	TUR (1 736)	UNK (463)	GEO (415)	ARM (336)	AZE (314)
13	3 641	42.1	53	AUT (1 532)	SUI (991)	GER (563)	UNK (301)	FRA (50)
14	3 570	61.1	44	SRB (2 181)	BUL (586)	UNK (309)	MNE (210)	MKD (111)
15	3 473	54.6	69	USA (1 897)	UNK (891)	CAN (466)	NGR (31)	IND (14)
16	3 139	57.8	45	CRO (1 813)	SLO (752)	BIH (260)	UNK (175)	ITA (62)
17	3 055	43.8	29	UKR (1 339)	RUS (1 201)	UNK (356)	GER (19)	FID (17)
18	2 917	90.5	29	GRE (2 641)	UNK (164)	CYP (47)	ALB (11)	RUS (8)
19	2 811	35.8	57	CUB (1 007)	UNK (780)	MEX (526)	VEN (416)	ESP (12)
20	2 328	22.7	59	EGY (528)	UNK (345)	IRQ (212)	JOR (193)	UAE (176)

TABLE A.1: Federation composition of the largest non-default latent pools. The five federation columns report the largest federations inside each pool, with player counts in parentheses. UNK denotes missing or unresolved federation metadata.

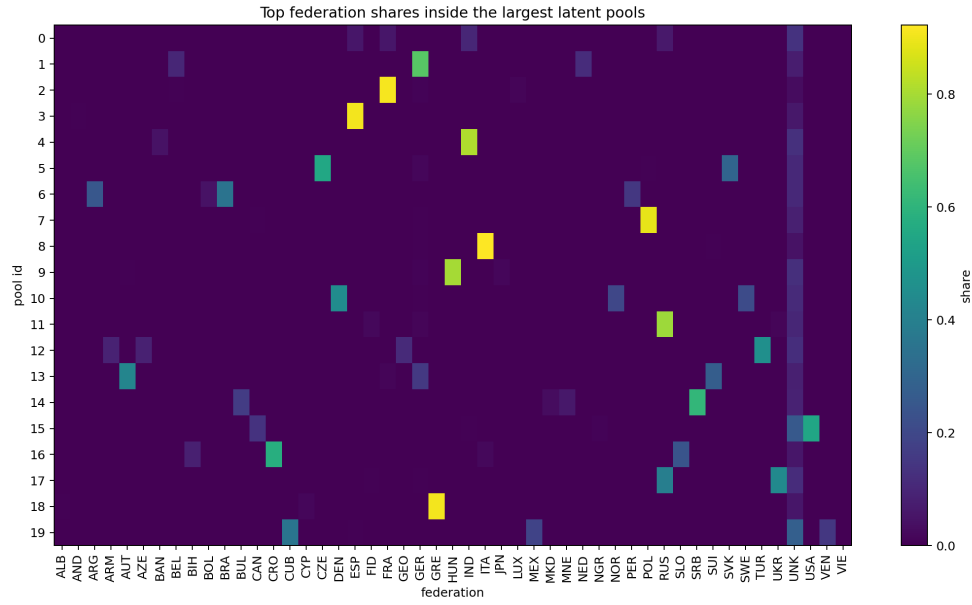


FIGURE A.1: Federation shares inside the largest non-default latent pools.

The pools were inferred from the game interaction graph, not from federation labels. Their federation composition shows that the interaction graph contains visible national and regional structure, but the correspondence is not exact.

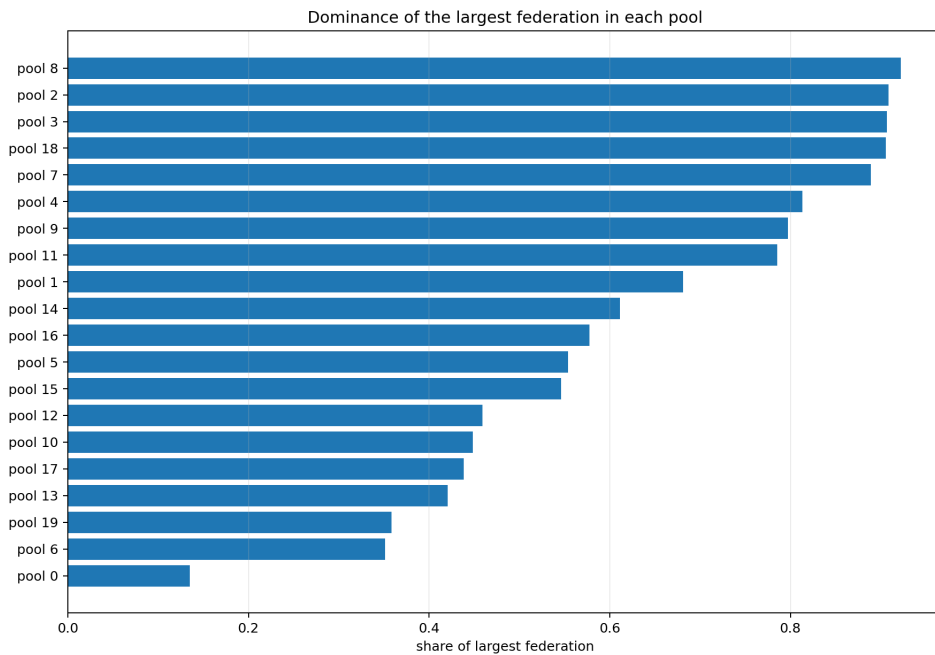


FIGURE A.2: Dominance of the largest federation in each major pool.

A.3 Federation-pool alignment

Table A.2 summarises the alignment between federations and pools using mutual information. When all players are included, the normalized mutual information is moderate because pool 0

dominates the population. After excluding the default pool, the alignment between federations and latent communities becomes much stronger.

Players used	Federations	Pools	MI [bits]	NMI sqrt	NMI min
520 260	160	57	1.006	0.304	0.523

TABLE A.2: Alignment between federations and learned pools. MI denotes mutual information. NMI sqrt is mutual information normalized by the square root of the federation and pool entropies. NMI min is mutual information normalized by the smaller of the two entropies. The table uses federations with at least 50 players.

The result suggests that federations are an important part of the interaction structure, but not the only one. A federation-based correction would impose a fixed administrative partition. The graph-derived pools also capture regional circuits, mixed environments and elite tournament structures when these appear in the game graph.

A.4 Elite player concentration

Table A.3 shows the distribution of the top 200 players by latent pool in the dataset snapshot. Pool 11 contains 83 of the top 200 players. Although many of its members are from Russia or neighbouring chess environments, the elite snapshot contains globally connected players from many federations. Pool 11 is therefore best read as an elite tournament environment rather than a purely national pool.

Pool	Top-200 players	Share [%]	Mean rating	U21
11	83	41.5	2661.7	1
4	15	7.5	2664.3	8
0	14	7.0	2629.5	10
27	13	6.5	2649.2	0
12	11	5.5	2623.0	0
15	10	5.0	2656.9	0
1	9	4.5	2660.7	0
17	7	3.5	2627.4	0
10	4	2.0	2646.5	0
33	4	2.0	2720.5	1

TABLE A.3: Distribution of the top 200 players across latent pools.

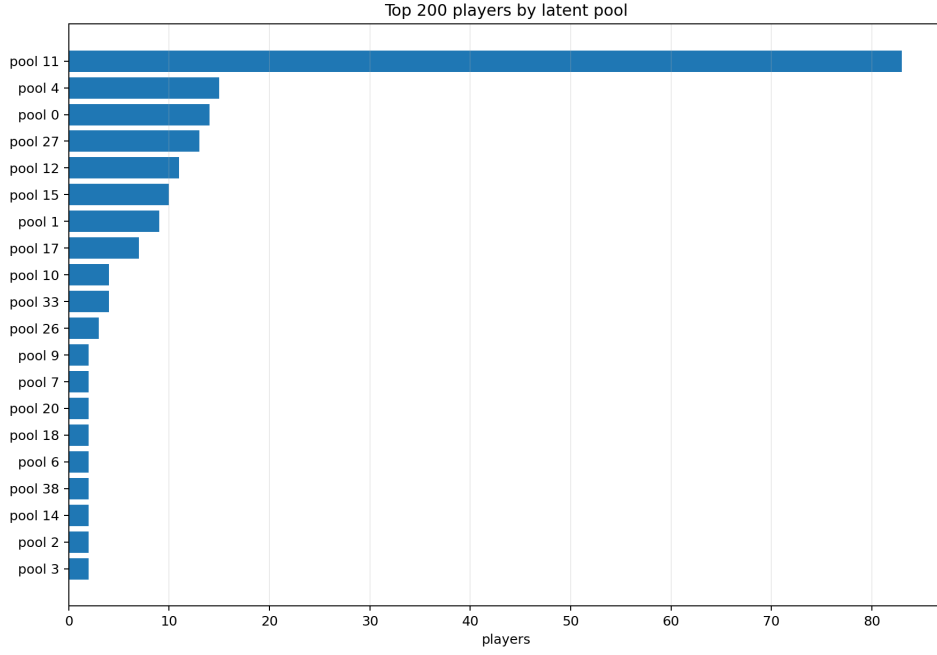


FIGURE A.3: Distribution of the top 200 players across latent pools.

Table A.4 shows the same top-200 snapshot by federation. The leading federations include USA, Russia, India, Ukraine and China. The table is a snapshot derived from the dataset, not a live ranking list.

Federation	Top-200 players	Share [%]	Mean rating	U21
USA	21	10.5	2672.2	3
RUS	18	9.0	2656.7	0
IND	18	9.0	2667.6	8
UKR	13	6.5	2626.5	0
CHN	12	6.0	2658.7	0
AZE	9	4.5	2646.4	1
FRA	9	4.5	2652.3	1
HUN	8	4.0	2660.4	0
ARM	7	3.5	2629.0	0
FID	6	3.0	2647.7	2

TABLE A.4: Distribution of the top 200 players across federations in the dataset snapshot.

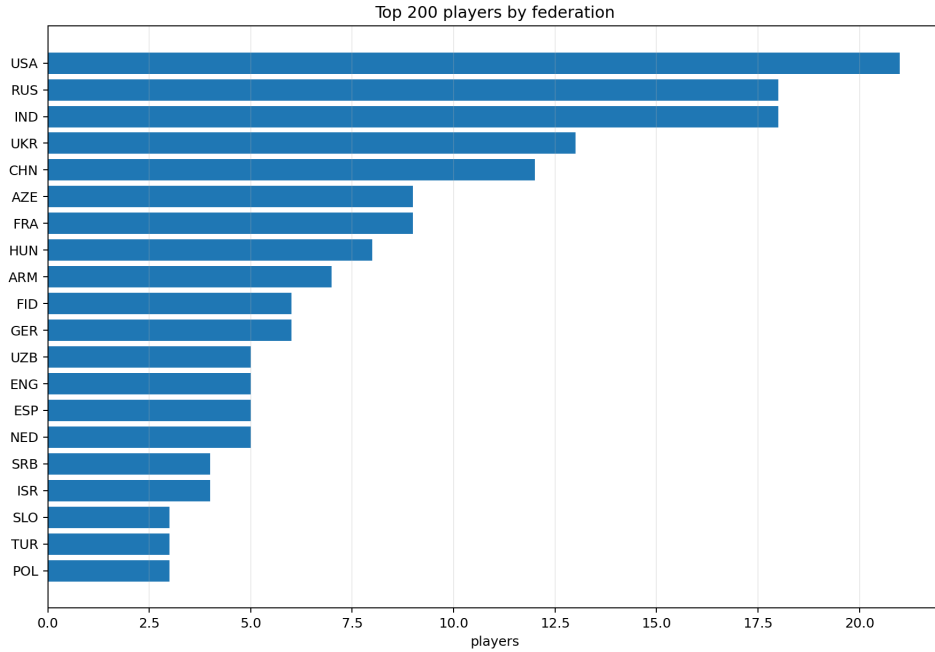


FIGURE A.4: Distribution of the top 200 players across federations in the dataset snapshot.

A.5 Junior and age profiles

Chess populations differ not only by geography but also by age. Table A.5 reports age and junior representation for large pools. Pool 4, dominated by India and Bangladesh, has a low median age and a large under-21 share. Pool 34, dominated by Sri Lanka in the detailed output, is also young. Several central and western European pools have much higher median ages.

Pool	Players	Known birth [%]	Median age	U21 among known [%]	Median rating
1	12 045	92.8	58.0	0.0	1960.0
2	10 719	96.8	52.0	0.6	1819.0
3	8 917	93.9	53.0	0.4	1862.5
4	7 391	87.1	29.5	3.2	1725.0
5	6 652	89.5	55.0	0.4	1911.0
6	4 938	89.2	40.0	0.5	1909.0
7	4 910	91.9	35.0	1.0	1818.0
8	4 794	95.6	57.0	0.2	1842.0
9	4 475	87.6	49.0	1.1	1893.0
10	4 177	89.1	59.0	0.1	1924.0
11	3 845	90.0	41.0	1.1	1965.0
12	3 781	87.8	32.0	1.4	1879.0
13	3 641	91.7	59.0	0.0	1973.0
14	3 570	91.3	55.0	0.2	1982.0
15	3 473	74.3	41.0	1.0	2003.0

TABLE A.5: Age and junior profile of the largest non-default pools. U21 denotes players younger than 21 among players with known birth year.

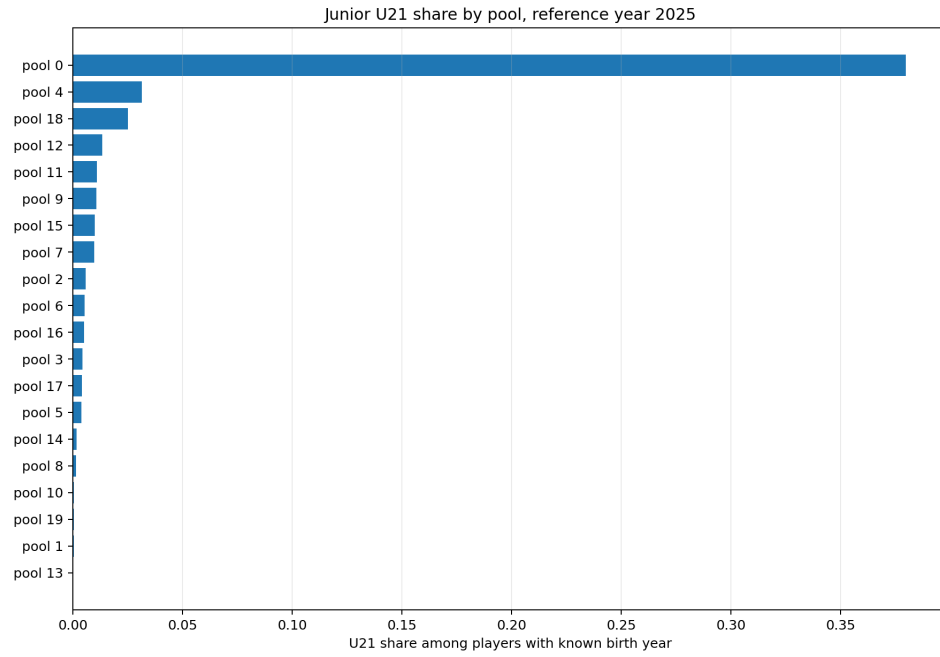


FIGURE A.5: Under-21 share by pool among players with known birth year.

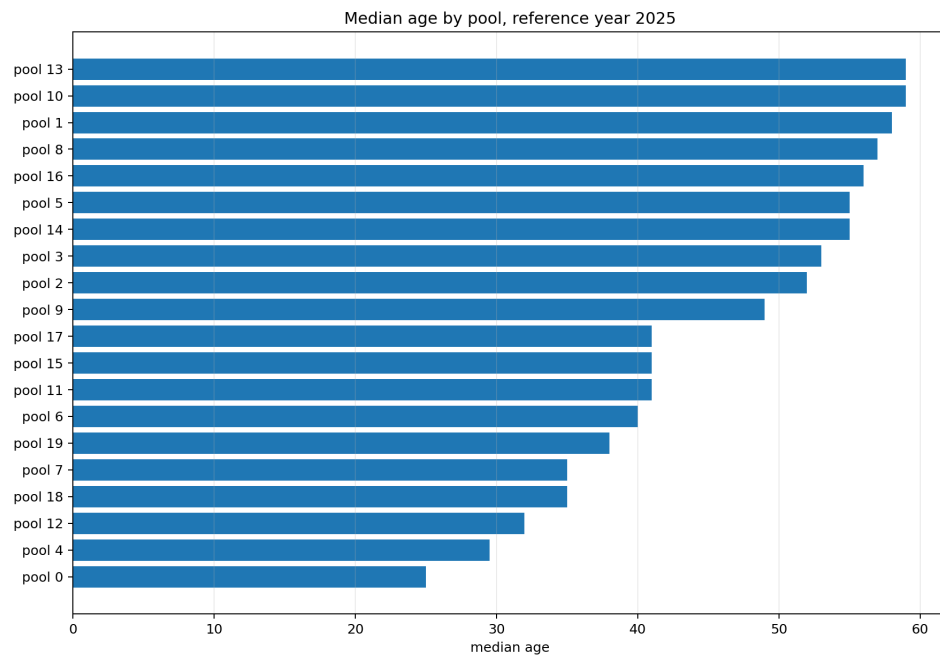


FIGURE A.6: Median age by pool among the largest non-default pools.

The contrast between young and mature playing environments is visible in both figures. The Indian-related pool and several Asian pools have substantially lower median age than many European pools. Such differences are relevant for rating prediction because younger pools contain more players whose playing strength may change rapidly during the observed period.

A.6 Junior participation over time

Table A.6 and Figure A.7 show the share of under-21 player-game entries over time. The share increases from about 27.7% in 2008 to about 49.7% in 2025. The share of active players who are under 21 also increases.

Year	Entries	U21 entries	U21 entry share [%]	U21 active share [%]	U21 avg. rating
2008	1 321 648	366 732	27.7	21.9	2000.3
2010	1 887 560	560 148	29.7	23.8	1909.3
2012	2 469 168	806 172	32.6	26.5	1820.5
2015	2 972 642	1 152 091	38.8	32.3	1673.6
2018	3 584 734	1 546 840	43.2	35.2	1564.2
2020	1 442 778	586 552	40.7	33.5	1560.6
2022	2 999 860	1 372 300	45.7	35.7	1534.3
2023	3 574 098	1 675 651	46.9	37.5	1519.9
2024	3 889 598	1 887 586	48.5	38.4	1667.7
2025	4 323 532	2 146 810	49.7	38.2	1725.9

TABLE A.6: Junior participation over time. U21 denotes players younger than 21 in a given year; entry share is measured by player-game entries.

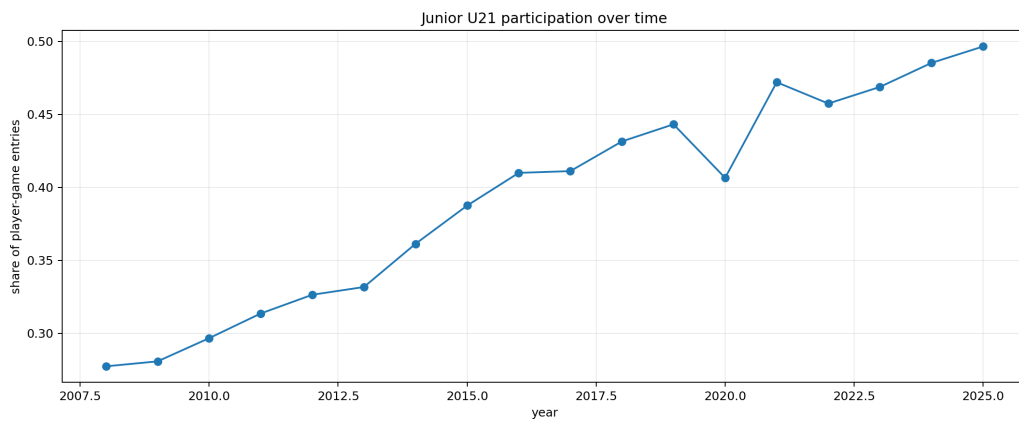


FIGURE A.7: Under-21 participation over time measured by player-game entries.

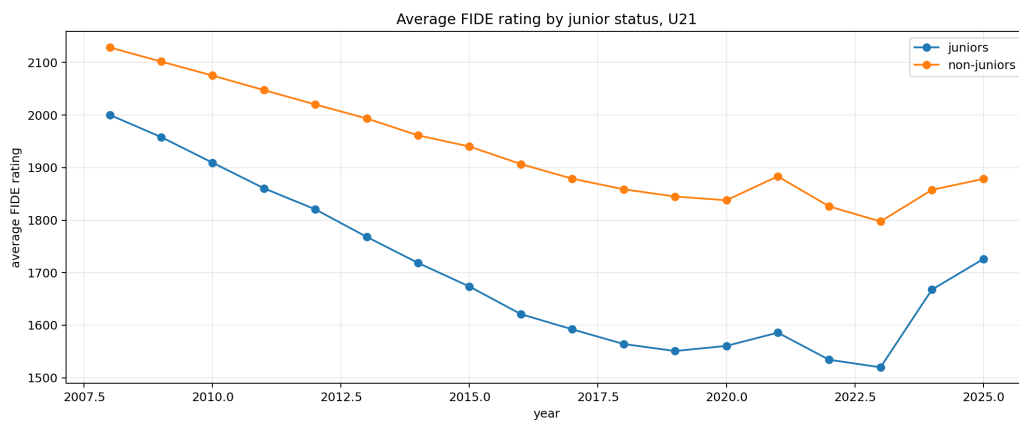


FIGURE A.8: Average FIDE rating of under-21 and non-under-21 player-game entries over time.

The increasing junior share provides context for the provisional-player experiments. A rating

system evaluated on recent FIDE data must handle a large and growing population of young players, many of whom are not well represented by a fixed initial rating.

A.7 Cross-pool interactions

Table A.7 reports both absolute and relative interaction volume. The relative value is computed as

$$1000 \cdot \frac{g_{ij}}{\sqrt{g_i g_j}},$$

where g_{ij} is the number of games between pools i and j , and g_i, g_j are the total game volumes of the two pools. This normalization reduces the bias toward large pools. It also reports the largest cross-pool interaction volumes after excluding default-pool and within-pool games. Many high-volume pairs are geographically plausible: German/Benelux with Austria/Switzerland/Germany, German/Benelux with France, Czech/Slovak with Poland, and elite or Russian-related pools with Ukrainian or Caucasus-related pools.

Pools	Games	Rel. volume [per 1000]	Avg. score low pool	Avg. rating diff
1–13	33 415	63.9	0.504	-4.4
1–2	30 446	34.8	0.515	21.5
5–7	16 933	34.0	0.454	43.8
1–5	16 043	21.0	0.477	-9.3
11–22	14 945	69.4	0.537	47.3
11–17	14 909	57.2	0.523	20.6
11–12	14 555	39.4	0.548	56.8
3–30	14 109	57.3	0.540	39.4
5–9	12 579	25.8	0.462	-23.0
1–10	11 869	18.8	0.506	7.8
1–11	11 567	20.2	0.341	-126.1
3–21	10 578	33.8	0.547	72.5
1–12	10 527	18.4	0.395	-60.0
14–16	10 241	40.9	0.546	26.3
2–3	10 011	12.4	0.471	-38.5

TABLE A.7: Largest absolute cross-pool interaction volumes since 2015, excluding default-pool and within-pool games. Relative volume normalises the number of games by the geometric mean of the total game volumes of the two pools.

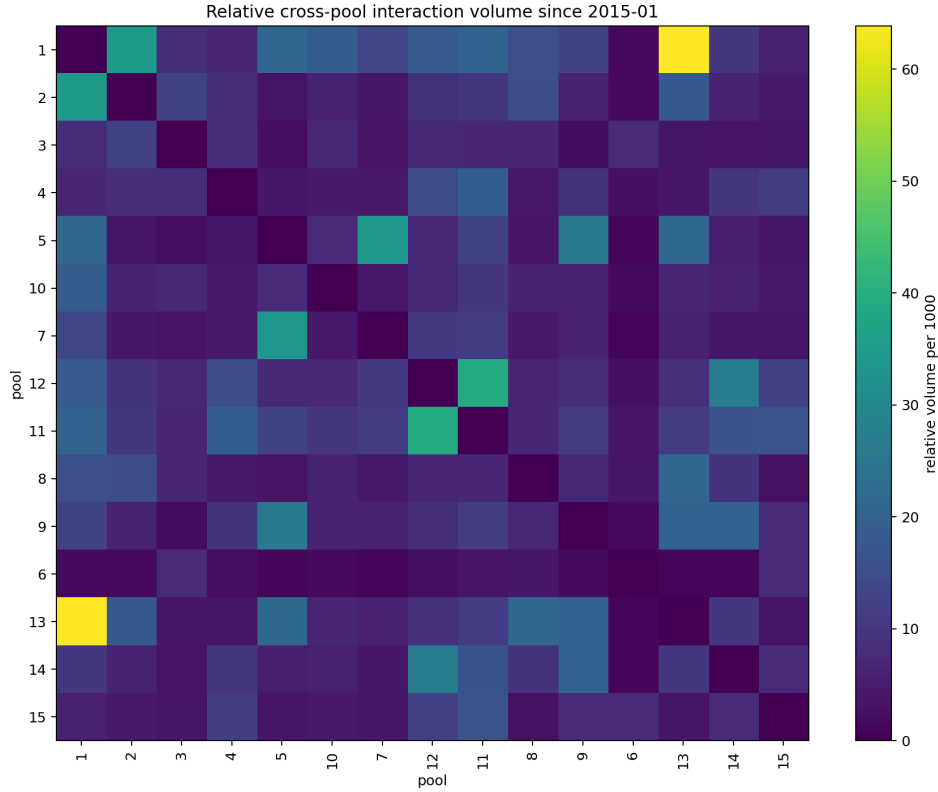


FIGURE A.9: Relative cross-pool interaction volume among the largest non-default pools since 2015.

The heatmap shows a block structure rather than uniform mixing. High-volume off-diagonal cells often correspond to neighbouring regions or elite bridges, while low-volume cells correspond to environments that rarely meet. The observed structure is neither fully mixed nor fully disconnected, which is consistent with the need for both global calibration and pool-level state.

A.8 Pool offsets and scale interpretation

Figure A.10 shows the largest learned pool offsets. Some offsets approach the allowed maximum. They should be read as internal predictive corrections within the fitted replay model. They are not public rating differences between federations or regions. Since the selected parameter vector has zero pool-centering strength, offsets can also absorb scale effects that would require separate normalisation in a public display system.

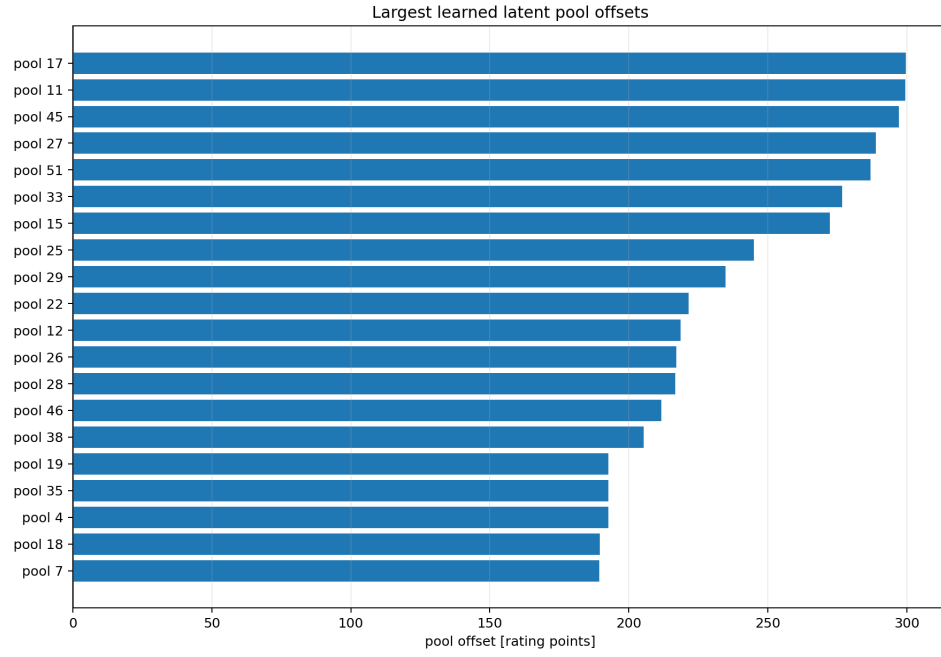


FIGURE A.10: Largest learned latent pool offsets. They are internal predictive corrections, not public rating differences.

Pairwise pool offsets are shown in Figure A.11. Their predictive role is smaller than the role of the main pool component, as indicated by the ablation study. They are therefore interpreted as secondary cross-pool corrections.

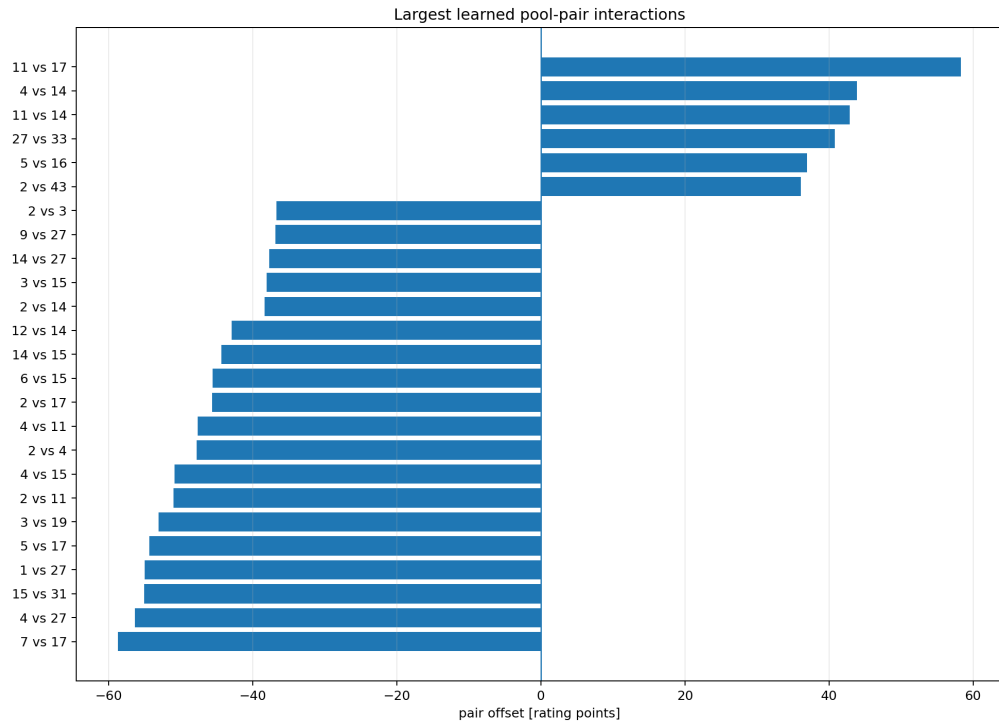


FIGURE A.11: Largest learned pairwise pool interactions.

A.9 Supplementary metadata diagnostic

The dataset contains a FIDE sex field for many players. The diagnostic in Figure A.12 is included as a metadata audit rather than as a central result. It can indicate whether a pool corresponds to a specialised tournament circuit, a metadata irregularity or a demographic pattern. The rating model itself does not use this field.

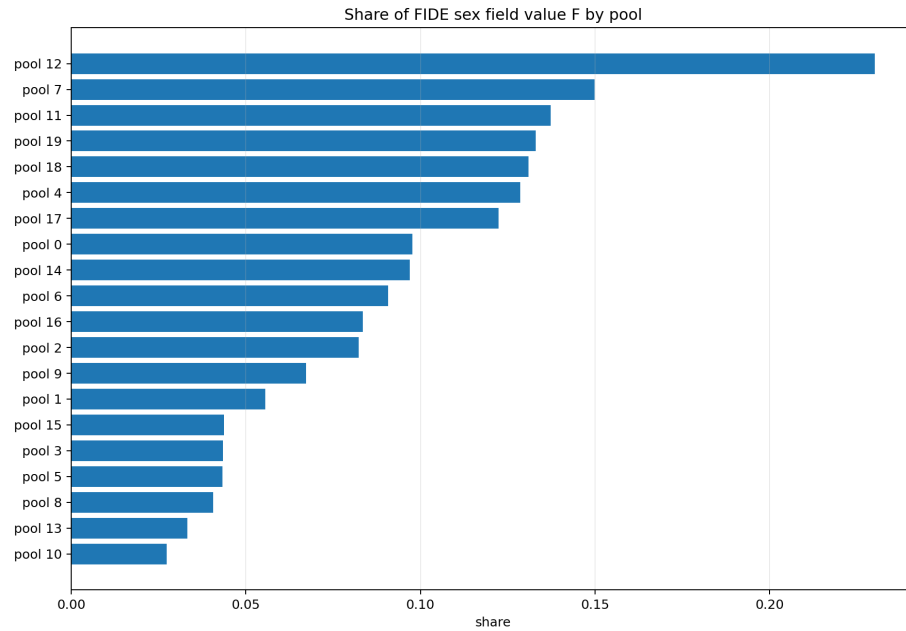


FIGURE A.12: Diagnostic share of the FIDE sex-field value F by pool.



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